ANALOG LAB · LECTURES 23 – 25 · REVISED EDITION

The CMOS Inverter from VTC to delay

Every idea explained from scratch, every derivation done line by line, every formula with a memory hook, and Razavi-style practice questions (each with its own figure and a full worked solution).

Part 0	Warm-up: MOSFET square law (with signs), the VTC, noise margins, the maths you need
Part 1 · L23	How the inverter works, the five regions A–E, deriving V_{IL} and V_{IH}
Part 2 · L24	Switching threshold V_{th}, sizing, the symmetric inverter, layout, the inverter as an amplifier
Part 3 · L25	Load capacitance, delay definitions, average-current and exact methods, ring oscillator, power
Part 4	One-page formula sheet, memory tricks, flashcards, exam playbook, capstone problem

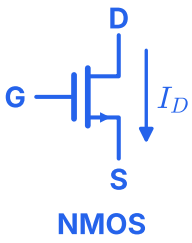
HOW TO USE THIS GUIDE

Blue boxes = results to memorise. **Green** = the intuition. **Red** = common mistakes. **Purple** = memory tricks. **Orange** = questions: try them first, then turn the page to the full solutions.

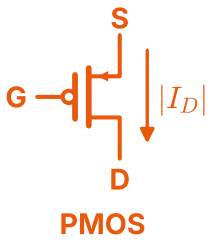
KEEP THIS PAGE OPEN

Reference card

The only device equations used in this whole guide. Everything else is built from these.



ON when the gate is HIGH: $V_{GS} > V_{T0,n}$ (>0)
current flows D → S



ON when the gate is LOW: $V_{GS} < V_{T0,p}$ (<0)
current flows S → D

	NMOS (ALL POSITIVE)	PMOS (ALL NEGATIVE)
Threshold	$V_{T0,n} > 0$	$V_{T0,p} < 0$
Strength	$k_n = \mu_n C_{ox} (W/L)_n$	$k_p = \mu_p C_{ox} (W/L)_p$
OFF	$V_{GS} < V_{T0,n},\ I_D = 0$	$V_{GS} > V_{T0,p},\ I_D = 0$
Linear (triode)	$V_{DS} < V_{GS} - V_{T0,n}$	$V_{DS} > V_{GS} - V_{T0,p}$
Saturation	$V_{DS} \geq V_{GS} - V_{T0,n}$	$V_{DS} \leq V_{GS} - V_{T0,p}$

THE TWO CURRENT FORMULAS (SAME FOR BOTH DEVICES)

$$\text{Linear: } I_D = \frac{k}{2}\left[2\left(V_{GS} - V_{T0}\right)V_{DS} - V_{DS}^2\right] \qquad \text{Saturation: } I_D = \frac{k}{2}\left(V_{GS} - V_{T0}\right)^2$$

INVERTER TRANSLATIONS (WRITE THESE FIRST IN EVERY QUESTION)

$$V_{GS,n} = V_{in} \qquad V_{DS,n} = V_{out} \qquad V_{GS,p} = V_{in} - V_{DD} \qquad V_{DS,p} = V_{out} - V_{DD} \qquad k_R = \frac{k_n}{k_p}$$

PMOS WITHOUT SIGN HEADACHES

Work in magnitudes. The PMOS is ON when $|V_{GS}| > |V_{T0,p}|$ and saturated when $|V_{DS}| \geq |V_{GS}| - |V_{T0,p}|$. That is **exactly the NMOS rule**, so you only ever remember one rule.

PART 0 · WARM-UP

Prerequisites

Everything Lectures 23–25 assume you already know, in plain words.

- ✓ Which region a transistor is in, in 3 checks
- ✓ What a VTC is and its five key numbers
- ✓ Implicit differentiation (used in L23)
- ✓ Why the PMOS is "the NMOS upside down"
- ✓ What a noise margin protects you from
- ✓ The one integral used in L25

0.1 The MOSFET in one picture

A MOSFET is a **voltage-controlled valve**. The gate voltage decides *if* current can flow. The drain voltage decides *how the valve behaves*:

- **OFF:** gate not above threshold, so no channel and no current.
- **Linear (triode):** small V_{DS}. The channel behaves like a **resistor**: more V_{DS}, more current.
- **Saturation:** large V_{DS}. The channel "pinches off" and the current stops growing. The device behaves like a **current source**.

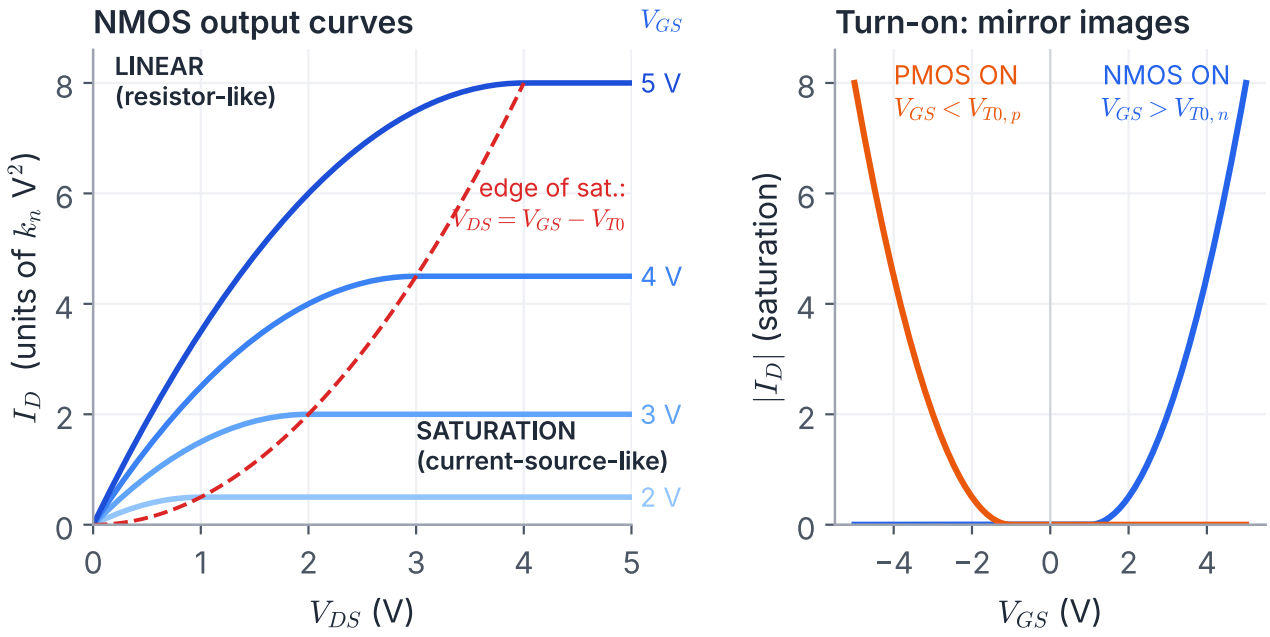


Fig. 0.1 Left: each curve rises like a resistor (linear), then goes flat like a current source (saturation). The dashed red parabola is the border between them. Right: the NMOS turns on for positive V_{GS}, the PMOS for negative V_{GS}, a perfect mirror image.

Finding the region: the 3-check recipe

1. **Is it ON?** NMOS: $V_{GS} > V_{T0,n}$. PMOS: $|V_{GS}| > |V_{T0,p}|$. If not, $I_D = 0$, stop.

2. **Compute the overdrive** $V_{ov} = V_{GS} - V_{T0}$ (use magnitudes for the PMOS).

3. **Compare $|V_{DS}|$ with $|V_{ov}|$.** $|V_{DS}| \geq |V_{ov}|$ → **saturation**. $|V_{DS}| < |V_{ov}|$ → **linear**.

THE PMOS INEQUALITY FLIP

With signed PMOS voltages, every "$\geq$" of the NMOS becomes "$\leq$" because everything is negative. If you ever feel unsure, switch to magnitudes (reference card) and the rule is identical to the NMOS.

0.2 The VTC and its five numbers

The **voltage transfer characteristic** (VTC) is simply the graph of V_{out} against V_{in}. For an inverter it starts high, drops, and ends low. Five numbers describe it:

SYMBOL	NAME	HOW TO FIND IT ON THE GRAPH
V_{OH}	output high	the flat top (input = 0)
V_{OL}	output low	the flat bottom (input = V_{DD})
V_{IL}	input low (max)	first point where slope $= -1$
V_{IH}	input high (min)	second point where slope $= -1$
V_{th}	switching threshold	where the VTC crosses the line $V_{out} = V_{in}$

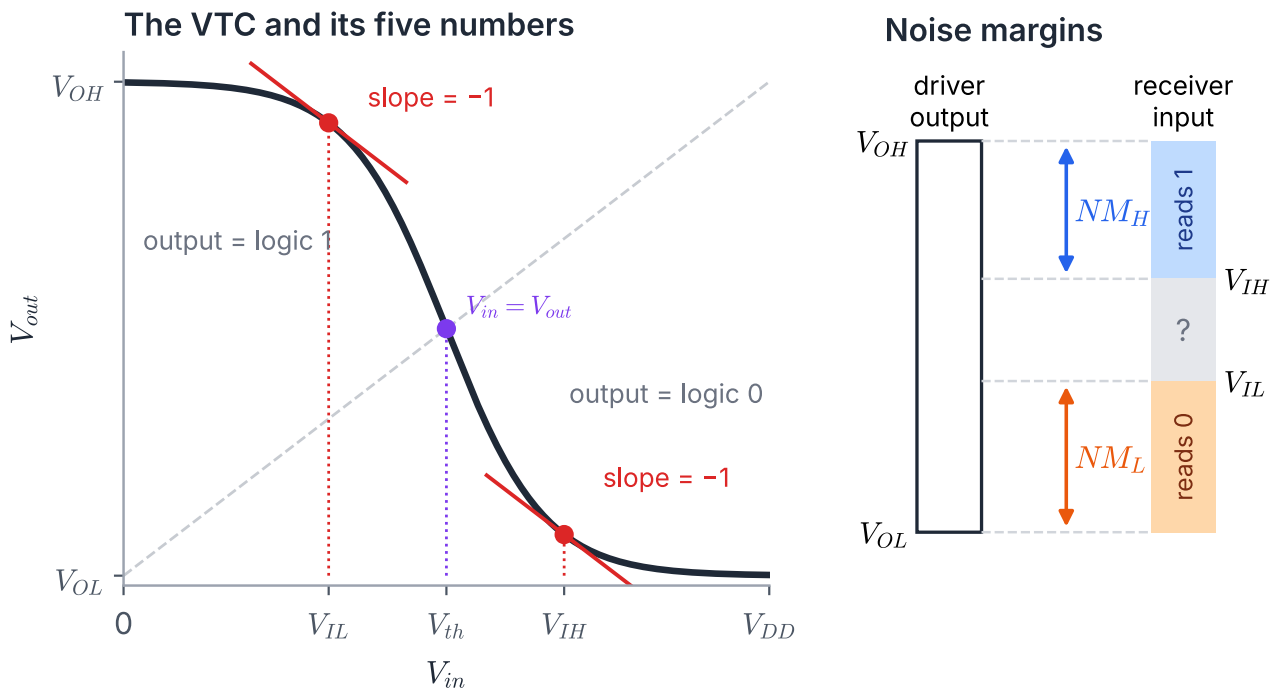


Fig. 0.2 Left: the five numbers on a VTC. Right: noise margins. The driver promises at least V_{OH} for a "1"; the receiver accepts anything above V_{IH} as a "1". The gap between them is the noise a "1" can survive. The grey "?" band is forbidden: the receiver cannot decide.

MEMORISE

$$NM_L = V_{IL} - V_{OL} \qquad NM_H = V_{OH} - V_{IH}$$

"Low margin uses the lows, high margin uses the highs."

WHY EXACTLY SLOPE = −1?

Noise Δ on the input appears at the output as $|\text{slope}| \cdot \Delta$. Where $|\text{slope}| < 1$ the noise **shrinks** (safe: a valid logic level). Where $|\text{slope}| > 1$ it **grows** (unsafe: the transition zone). The two slope = −1 points are exactly the borders between "safe" and "unsafe", so they define V_{IL} and V_{IH}.

0.3 The two pieces of maths you need

(a) Implicit differentiation (Lecture 23)

On the VTC, V_{out} is a function of V_{in}. Differentiate an equation with respect to V_{in}. Then every V_{out} term gets a factor $\frac{dV_{out}}{dV_{in}}$ (chain rule). This is the same as y in $\frac{d}{dx}(y^2) = 2y\,y'$:

$$\frac{d}{dV_{in}}(V_{out} - V_{DD})^2 = 2(V_{out} - V_{DD})\frac{dV_{out}}{dV_{in}}, \qquad \frac{d}{dV_{in}}\Big[(V_{in} - a)V_{out}\Big] = V_{out} + (V_{in} - a)\frac{dV_{out}}{dV_{in}}$$

Then you put $\frac{dV_{out}}{dV_{in}} = -1$ and the derivative disappears.

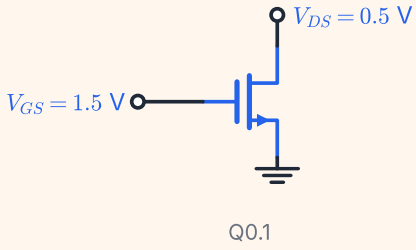
(b) One integral (Lecture 25)

$$\int \frac{dV}{V(a-V)} = \frac{1}{a}\ln\frac{V}{a-V} \qquad \text{because} \qquad \frac{1}{V(a-V)} = \frac{1}{a}\left[\frac{1}{V} + \frac{1}{a-V}\right] \quad \text{(partial fractions)}$$

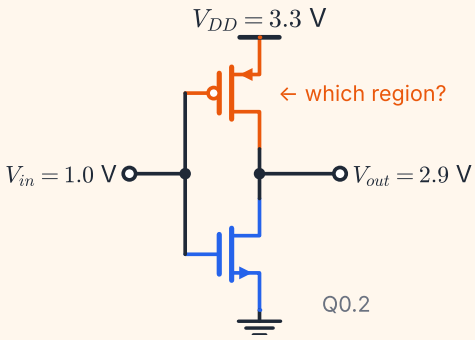
YOUR TURN · PART 0 (SOLUTIONS ON THE NEXT PAGE)

0.1 An NMOS has $V_{T0,n} = 0.6$ V and $k_n = 200\ \mu$A/V². It is biased at $V_{GS} = 1.5$ V, $V_{DS} = 0.5$ V. Which region, and what is I_D?

0.2 In an inverter with $V_{DD} = 3.3$ V, the PMOS has $V_{T0,p} = -0.7$ V and $k_p = 80\ \mu$A/V². With $V_{in} = 1.0$ V and $V_{out} = 2.9$ V, find $V_{GS,p}$, $V_{DS,p}$, the region and $|I_D|$.



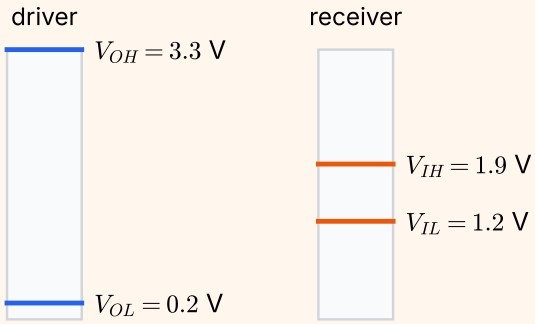
Q0.1



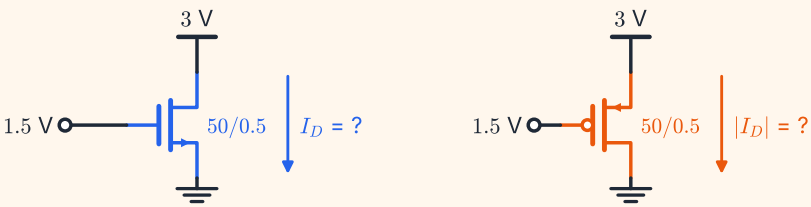
Q0.2

0.3 Differentiate $2\,(V_{in} - V_{DD} - V_{T0,p})(V_{out} - V_{DD})$ with respect to V_{in}.

0.4 A driver gives $V_{OH} = 3.3$ V, $V_{OL} = 0.2$ V; the receiver has $V_{IL} = 1.2$ V, $V_{IH} = 1.9$ V. Find both noise margins. Which logic level is better protected?



0.5 Razavi Prob. 2.1, with numbers Using Razavi's Table 2.1 data ($\mu_n C_{ox} = 134.3$, $\mu_p C_{ox} = 38.4\ \mu$A/V², $V_{THN} = 0.7$, $V_{THP} = -0.8$ V, $\lambda = 0$), find $|I_D|$ for an NMOS and a PMOS, both $W/L = 50/0.5$, with $|V_{GS}| = 1.5$ V and $|V_{DS}| = 3$ V. What does the ratio tell you?



Solutions · Part 0

0.1 ON, since $1.5 > 0.6$. Overdrive $V_{ov} = 0.9$ V. $V_{DS} = 0.5 < 0.9$ → **linear**.

$$I_D = \tfrac{200\mu}{2}\left[2(0.9)(0.5) - 0.5^2\right] = 100\mu \times 0.65 = \mathbf{65\ \mu A}$$

0.2 $V_{GS,p} = 1.0 - 3.3 = -2.3$ V, $V_{DS,p} = 2.9 - 3.3 = -0.4$ V. In magnitudes: $|V_{ov}| = 2.3 - 0.7 = 1.6$ V and $|V_{DS}| = 0.4 < 1.6$ → **linear**.

$$|I_D| = \tfrac{80\mu}{2}\left[2(1.6)(0.4) - 0.4^2\right] = 40\mu \times 1.12 = \mathbf{44.8\ \mu A}$$

0.3 Product rule; only V_{out} carries the chain-rule factor:

$$2\left[(V_{in} - V_{DD} - V_{T0,p})\frac{dV_{out}}{dV_{in}} + (V_{out} - V_{DD})\right]$$

This is exactly the middle term you meet in the V_{IL} derivation (§1.5).

0.4 $NM_L = 1.2 - 0.2 = \mathbf{1.0\ V}$, $NM_H = 3.3 - 1.9 = \mathbf{1.4\ V}$. The **logic 1** is better protected.

0.5 Both are saturated ($|V_{DS}| = 3$ V is larger than both overdrives).
NMOS: $k_n = 134.3\mu \times 100 = 13.43$ mA/V², $I_D = \frac{13.43\text{m}}{2}(1.5 - 0.7)^2 = \mathbf{4.30\ mA}$.
PMOS: $k_p = 3.84$ mA/V², $|I_D| = \frac{3.84\text{m}}{2}(1.5 - 0.8)^2 = \mathbf{0.94\ mA}$.
The same-size PMOS gives about **4.6× less current** (3.5× from mobility, the rest from its larger threshold).
This is the whole reason the PMOS in an inverter is drawn wider (Part 2).

PART 1 · LECTURE 23

The CMOS inverter VTC

Two complementary switches, five regions, and the slope = −1 derivations.

> **Big idea: the NMOS and PMOS are in a tug-of-war over V_{out}; the input decides who is stronger.**

- ✓ Read the four terminal voltages off the circuit
- ✓ Name the region of each transistor in A–E
- ✓ Check your answer with a one-line shortcut
- ✓ Draw the two "saturation boundary" lines
- ✓ Derive V_{IL} and V_{IH} step by step
- ✓ Explain why $V_{OH} = V_{DD}$ and $V_{OL} = 0$ exactly

1.1 The circuit

Two transistors, gates joined (that is the input), drains joined (that is the output). The PMOS sits on top and can **pull the output up** to V_{DD}; the NMOS sits underneath and can **pull it down** to ground. The same current flows through both: $i_{D,p} = i_{D,n}$ whenever nothing else draws current from the output.

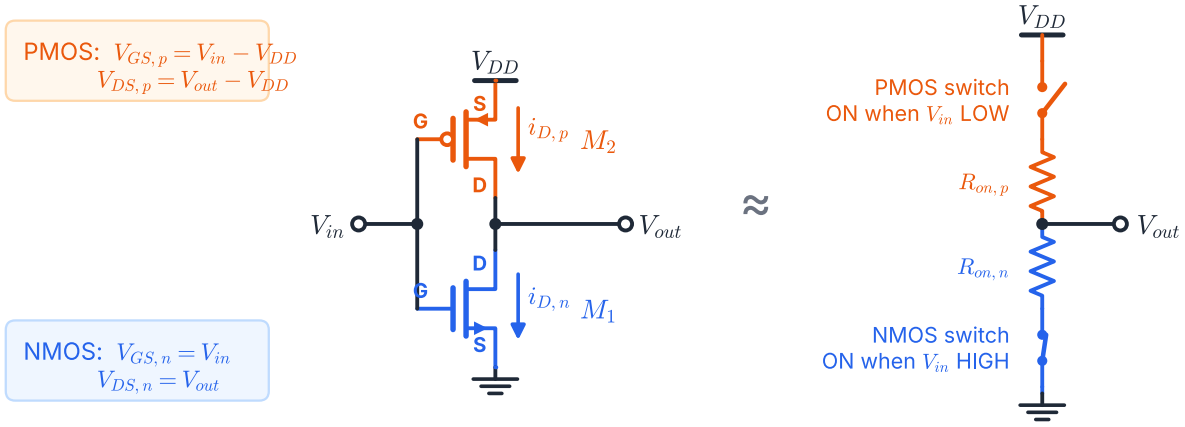


Fig. 1.1 Left: the CMOS inverter with gate (G), drain (D) and source (S) marked. The PMOS source is at V_{DD}; the NMOS source is at ground. Arrows show the drain current flowing from V_{DD}, through both devices, to ground. Right: the lecture's "non-ideal switch" model: each transistor is a switch with an on-resistance, and in steady state only one switch is closed.

Reading the terminal voltages off the figure is **step one of every CMOS question**:

$$V_{GS,n} = V_{in}, \qquad V_{DS,n} = V_{out}, \qquad V_{GS,p} = V_{in} - V_{DD} = -(V_{DD} - V_{in}), \qquad V_{DS,p} = V_{out} - V_{DD} = -(V_{DD} - V_{out})$$

1.2 The story as V_{in} goes from 0 to V_{DD}

Remember each region as a tiny picture: a transistor that is **off** is an **open switch**, a **linear** one is a **resistor**, a **saturated** one is a **current source**.

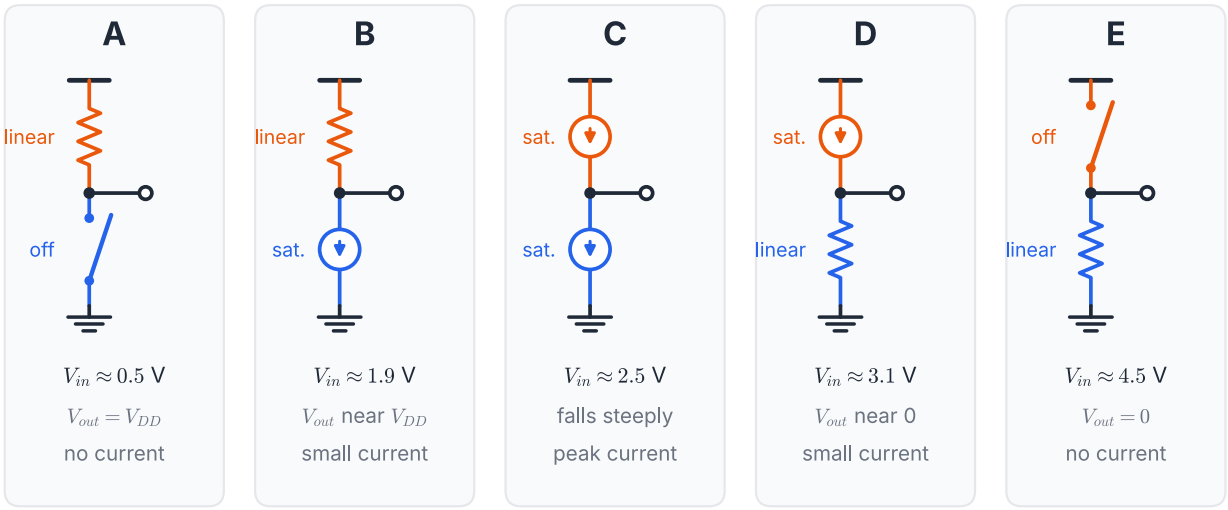


Fig. 1.2 The five snapshots A–E. In A the NMOS is open, so the PMOS "resistor" just connects the output to V_{DD}. In B the NMOS wakes up as a weak current source and drags the output slightly down. In C both are current sources fighting each other, so the output falls steeply. D and E are the mirror image.

1.3 The two saturation boundaries

Each transistor's "edge of saturation" is a **straight line** on the (V_{in}, V_{out}) plane. Drawing them tells you the regions without any algebra.

1. **NMOS saturated** needs $V_{DS,n} \geq V_{GS,n} - V_{T0,n}$, i.e.

$$\boxed{V_{out} \geq V_{in} - V_{T0,n}} \qquad \text{(NMOS saturated ABOVE this line)}$$

2. **PMOS saturated** needs $V_{DS,p} \leq V_{GS,p} - V_{T0,p}$:

$$V_{out} - \cancel{V_{DD}} \leq V_{in} - \cancel{V_{DD}} - V_{T0,p} \Rightarrow \boxed{V_{out} \leq V_{in} - V_{T0,p}} \qquad \text{(PMOS saturated BELOW this line)}$$

Both lines have slope 1. The NMOS line starts at $V_{in} = V_{T0,n}$ on the axis; the PMOS line is the same line shifted up by $|V_{T0,p}|$.

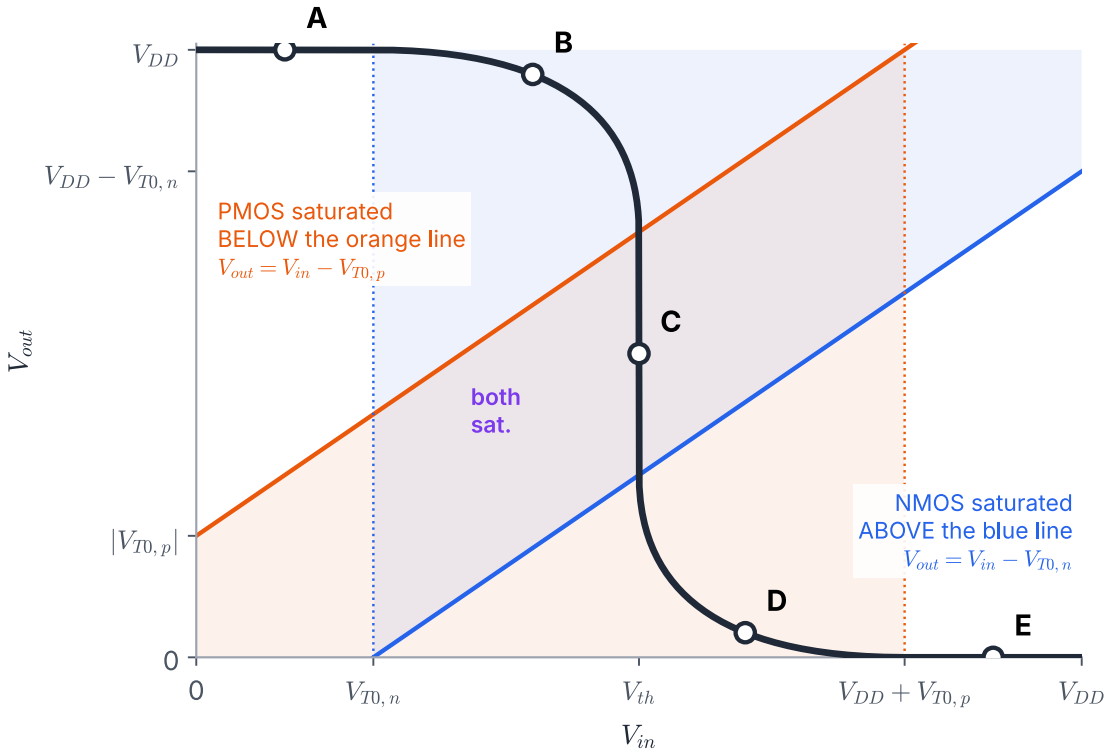


Fig. 1.3 The lecture's hatched map, computed exactly ($V_{DD} = 5$ V, $V_T = \pm 1$ V, $k_R = 1$). Blue area: NMOS saturated. Orange area: PMOS saturated. Where both overlap (the band between the lines), both are saturated, and that is where the VTC falls through C.

1.4 The five regions

POINT	INPUT RANGE	NMOS	PMOS	WHAT YOU GET
A	$V_{in} < V_{T0,n}$	off	linear	$V_{OH} = V_{DD}$
B	$V_{T0,n} \le V_{in} < V_{th}$	saturated	linear	V_{IL} lies here
C	$V_{in} = V_{th}$	saturated	saturated	V_{th} (Lecture 24)
D	$V_{th} < V_{in} \le V_{DD} + V_{T0,p}$	linear	saturated	V_{IH} lies here
E	$V_{in} > V_{DD} + V_{T0,p}$	linear	off	$V_{OL} = 0$

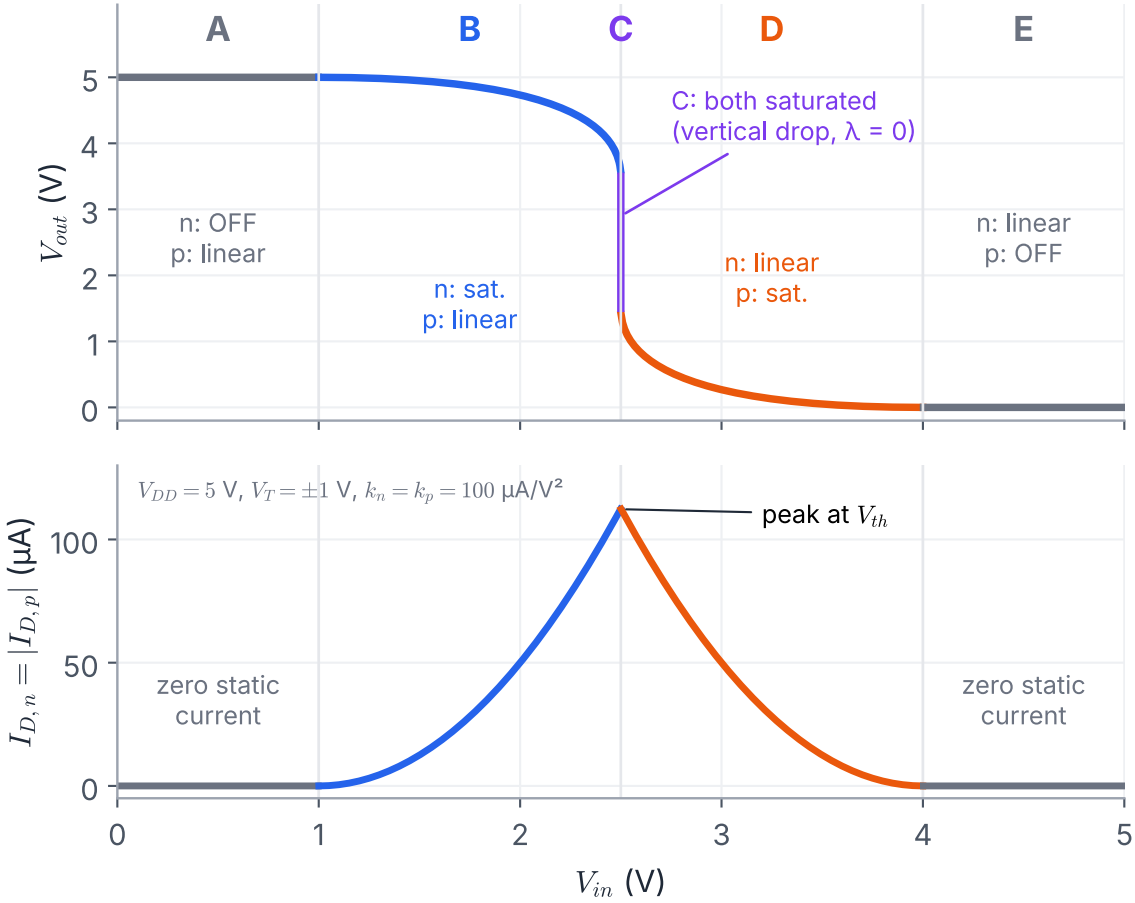


Fig. 1.4 Top: the VTC coloured by region. Bottom: the current through the inverter. It is **zero at both ends**, so a CMOS inverter uses no static power. It peaks at V_{th}, where both devices conduct hard.

MEMORY TRICK: "WEAKLY ON = SATURATED"

The transistor that is **only just on** (near its threshold) is saturated. The transistor that is **strongly on** holds the output near its own rail, so it is linear. In B the NMOS is only just on. In D the PMOS is only just on.

CHANT IT

NMOS first: **"Off-Lin, Sat-Lin, Sat-Sat, Lin-Sat, Lin-Off"**. The middle three read the same backwards around C.

WHY $V_{OH} = V_{DD}$ AND $V_{OL} = 0$ EXACTLY

At A the NMOS is off, so the current is zero. A linear PMOS carrying zero current has zero voltage across it (a resistor with no current), so $V_{out} = V_{DD}$. At E the same argument with roles swapped gives $V_{out} = 0$. A resistor-load inverter cannot do this: its NMOS is still conducting at $V_{in} = V_{OH}$, so $V_{OL} > 0$.

1.5 Deriving V_{IL} (point B: NMOS sat., PMOS linear)

Always use the same four moves:

equate currents → differentiate → put slope = −1 → solve

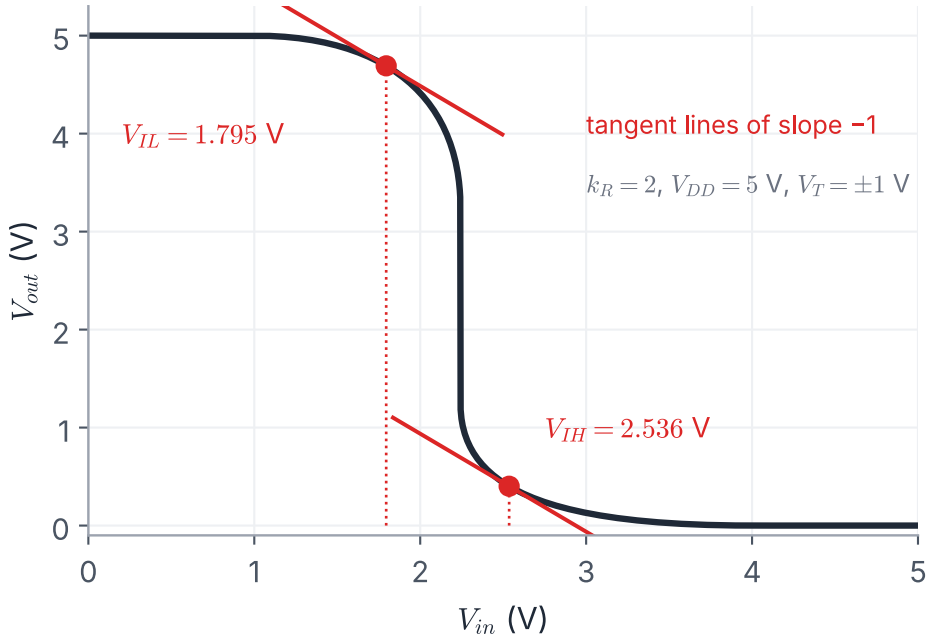


Fig. 1.5 What we are hunting: the two points where a line of slope −1 just touches the VTC (numbers from Worked example 23.1).

1 **Equate the currents.** Write NMOS saturation formula = PMOS linear formula. Divide both sides by $k_p/2$:

$$k_R(V_{in} - V_{T0,n})^2 = 2\underbrace{(V_{in} - V_{DD} - V_{T0,p})}_{V_{GS,p} - V_{T0,p}}\underbrace{(V_{out} - V_{DD})}_{V_{DS,p}} - (V_{out} - V_{DD})^2 \qquad ①$$

2 **Differentiate with respect to V_{in}** (chain rule, §0.3) and cancel the common 2:

$$k_R(V_{in} - V_{T0,n}) = (V_{in} - V_{DD} - V_{T0,p})\frac{dV_{out}}{dV_{in}} + (V_{out} - V_{DD}) - (V_{out} - V_{DD})\frac{dV_{out}}{dV_{in}}$$

3 **Put $V_{in} = V_{IL}$ and $\frac{dV_{out}}{dV_{in}} = -1$:**

$$k_R(V_{IL} - V_{T0,n}) = -V_{IL} + V_{DD} + V_{T0,p} + 2V_{out} - 2V_{DD}$$

4 **Collect the V_{IL} terms** on the left as $(1 + k_R)V_{IL}$:

$$\boxed{V_{IL} = \frac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R}} \qquad ②$$

5 **Solve ① and ② together.** Rearrange ② for V_{out}, substitute into ①, and you get a quadratic in V_{IL}. Keep the root between $V_{T0,n}$ and V_{th}.

$$V_{out}\big|_{V_{IL}} = \tfrac{1}{2}\left[(1 + k_R)V_{IL} + V_{DD} - V_{T0,p} - k_R V_{T0,n}\right]$$

1.6 Deriving V_{IH} (point D: NMOS linear, PMOS sat.)

1. **Equate the currents** with the roles swapped:

$$k_R\left[2(V_{in}-V_{T0,n})V_{out}-V_{out}^2\right]=(V_{in}-V_{DD}-V_{T0,p})^2 \qquad ③$$

2. **Differentiate** and cancel the 2s:

$$k_R\left[(V_{in}-V_{T0,n})\frac{dV_{out}}{dV_{in}}+V_{out}-V_{out}\frac{dV_{out}}{dV_{in}}\right]=V_{in}-V_{DD}-V_{T0,p}$$

3. **Put slope** $=-1$: $k_R\left[-V_{IH}+V_{T0,n}+2V_{out}\right]=V_{IH}-V_{DD}-V_{T0,p}$

4. **Collect:**

$$\boxed{V_{IH}=\frac{V_{DD}+V_{T0,p}+k_R(2V_{out}+V_{T0,n})}{1+k_R}} \qquad ④$$

5. **Solve ③ and ④ together** for V_{IH} and $V_{out}\big|_{V_{IH}}$.

TELLING ② AND ④ APART

Both have $1+k_R$ underneath and $2V_{out}$ on top. In ② (V_{IL}) k_R multiplies **only** $V_{T0,n}$. In ④ (V_{IH}) k_R multiplies $(2V_{out}+V_{T0,n})$, because at D the NMOS is linear and its current depends on V_{out}.

WORKED EXAMPLE 23.1 · THE LECTURE METHOD, START TO FINISH

$V_{DD}=5$ V, $V_{T0,n}=1$ V, $V_{T0,p}=-1$ V, $k_n=200$, $k_p=100$ μA/V², so $k_R=2$.

V_{IL}. From ②: $V_{IL}=\frac{2V_{out}-1-5+2}{3}$, so $V_{out}=1.5V_{IL}+2$ and $V_{out}-5=1.5V_{IL}-3$. Write $x=V_{IL}$ and substitute into ①:

$$2(x-1)^2=2(x-4)(1.5x-3)-(1.5x-3)^2=(1.5x-3)(0.5x-5)$$

$$2x^2-4x+2=0.75x^2-9x+15\;\Rightarrow\;1.25x^2+5x-13=0\;\Rightarrow\;x=\frac{-5+\sqrt{90}}{2.5}=\mathbf{1.795\ V}$$

with $V_{out}=1.5(1.795)+2=4.692$ V.

V_{IH}. From ④: $3V_{IH}=4+2(2V_{out}+1)$, so $V_{out}=0.75V_{IH}-1.5$. Substitute into ③:

$$2\left[2(x-1)(0.75x-1.5)-(0.75x-1.5)^2\right]=(x-4)^2\;\Rightarrow\;0.875x^2+3.5x-14.5=0\;\Rightarrow\;x=\mathbf{2.536\ V}$$

Margins: $NM_L=1.795$ V, $NM_H=5-2.536=2.464$ V. The stronger NMOS ($k_R>1$) pulls both V_{IL} and V_{IH} to the left.

SHORTCUT TO CHECK YOUR QUADRATIC

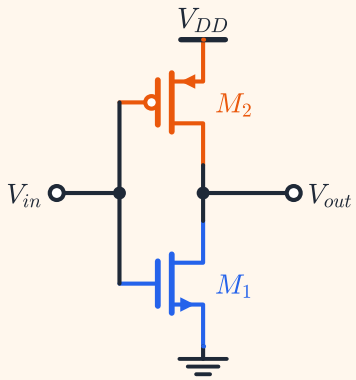
Eliminating V_{out} once and for all gives V_{IL}, V_{IH} and V_{th} in **one shape**:

$$V_x = \frac{V_{T0,n} + \rho\,(V_{DD} + V_{T0,p})}{1+\rho}, \qquad \rho_{IL} = \frac{2\sqrt{k_R(k_R+3)} - k_R}{k_R(k_R+4)}, \quad \rho_{th} = \frac{1}{\sqrt{k_R}}, \quad \rho_{IH} = \frac{1+2\sqrt{1+3k_R}}{3k_R}$$

Read it as a **weighted average** of $V_{T0,n}$ (NMOS turns on) and $V_{DD} + V_{T0,p}$ (PMOS turns off). For $k_R = 2$: $\rho_{IL} = (2\sqrt{10} - 2)/12 = 0.360$, so $V_{IL} = (1 + 0.360 \times 4)/1.360 = 1.795$ V ✓. For $k_R = 1$ you get $\rho = 3/5,\ 1,\ 5/3$, the symmetric results of Lecture 24. In an exam show the lecture method; use this to check.

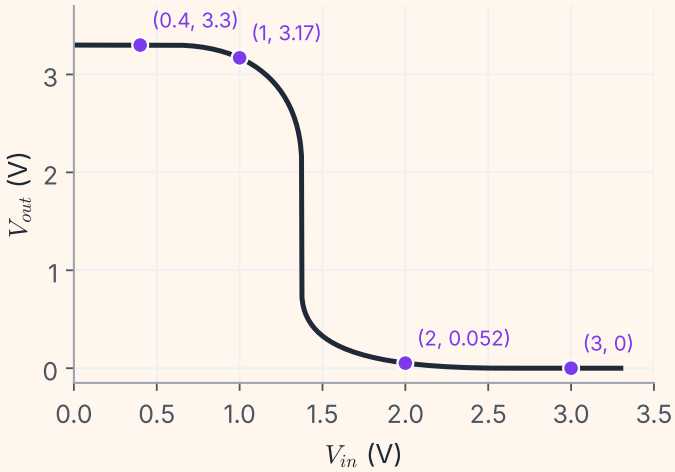
YOUR TURN · PART 1 (SOLUTIONS ON THE NEXT PAGE)

1.1 Inverter tutorial Q3 Find V_{OH}, V_{OL}, V_{IL}, V_{IH} and both noise margins.

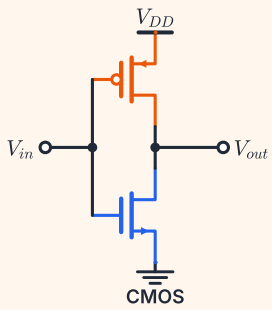


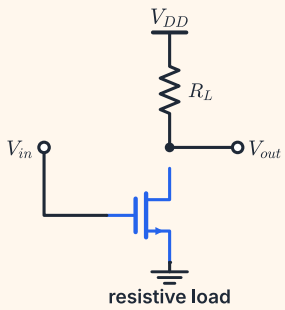
$V_{DD} = 3.3$ V
$V_{T0,\,n} = 0.6$ V, $V_{T0,\,p} = -0.7$ V
$k_n = 200$ µA/V², $k_p = 80$ µA/V²
Find $V_{OH}, V_{OL}, V_{IL}, V_{IH}$, NMs

1.2 For the inverter of 1.1, the four marked points lie on its VTC. Name the region of each transistor at each point, using only the boundary lines of §1.3.

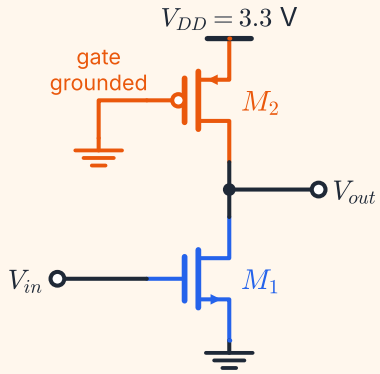


1.3 Razavi §3.2–3.3 idea Explain without equations why $V_{OL} = 0$ exactly in the CMOS inverter, but $V_{OL} > 0$ in the resistive-load inverter on the right.





1.4 Razavi Fig. 3.19(b) idea The PMOS gate is now **tied to ground** instead of the input ("pseudo-NMOS"). Find V_{OL} and the static power when $V_{in} = V_{DD}$. Why does a real CMOS inverter beat this?



$k_n = 200$ µA/V², $V_{T0,n} = 0.6$ V
$k_p = 80$ µA/V², $V_{T0,p} = -0.7$ V
Find V_{OL} and the power drawn when $V_{in} = V_{DD}$

Solutions · Part 1

1.1 $V_{OH} = 3.3$ V and $V_{OL} = 0$ V (CMOS rails). $k_R = 200/80 = 2.5$.
V_{IL}: ② gives $V_{out} = \frac{1}{2}[3.5V_{IL} + 3.3 + 0.7 - 1.5] = 1.75V_{IL} + 1.25$. Into ①: quadratic with root $V_{IL} = 1.065$ **V** ($V_{out} = 3.113$ V).
V_{IH}: ④ gives $3.5V_{IH} = 2.6 + 2.5(2V_{out} + 0.6)$, so $V_{out} = 0.7V_{IH} - 0.82$. Into ③: $V_{IH} = 1.553$ **V** ($V_{out} = 0.267$ V).
$NM_L = 1.065$ **V,** $NM_H = 3.3 - 1.553 = 1.747$ **V.** Shortcut check: $\rho_{IL} = 0.3025$ → $V_{IL} = (0.6 + 0.3025 \times 2.6)/1.3025 = 1.065$ ✓.

1.2 Boundaries: NMOS sat. if $V_{out} \geq V_{in} - 0.6$; PMOS sat. if $V_{out} \leq V_{in} + 0.7$.

(V_{IN}, V_{OUT})	NMOS	PMOS	POINT
(0.4, 3.3)	off ($0.4 < 0.6$)	linear ($3.3 > 1.1$)	A
(1.0, 3.170)	sat. ($3.17 \geq 0.4$)	linear ($3.17 > 1.7$)	B
(2.0, 0.052)	linear ($0.052 < 1.4$)	sat. ($0.052 \leq 2.7$)	D
(3.0, 0)	linear	off ($3.0 > 3.3 - 0.7$)	E

1.3 CMOS: at $V_{in} = V_{DD}$ the PMOS is **off**, so no current can flow anywhere. A linear NMOS with zero current has zero V_{DS}, so $V_{out} = 0$. Resistive load: the resistor is **always** connected to V_{DD}, so current flows and the output settles where $(V_{DD} - V_{out})/R_L = I_{D,n}$, which needs $V_{out} > 0$. The same argument at $V_{in} = 0$ gives $V_{OH} = V_{DD}$ for CMOS.

1.4 With its gate at 0 V the PMOS is always on: $|V_{GS,p}| = 3.3$ V, $|V_{ov}| = 2.6$ V. Guess PMOS saturated (output low): $|I_p| = \frac{80\mu}{2}(2.6)^2 = 270.4\ \mu$A. NMOS ($V_{GS} = 3.3$, $V_{ov} = 2.7$) is linear:

$$100\mu\left[2(2.7)V - V^2\right] = 270.4\mu \;\Rightarrow\; V^2 - 5.4V + 2.704 = 0 \;\Rightarrow\; V_{OL} = \mathbf{0.559\ V}$$

Check: PMOS $|V_{DS}| = 2.74 \geq 2.6$ ✓ saturated. Static power $= 3.3 \times 270.4\mu =$ **0.89 mW**, burned **all the time** the output is low. In real CMOS the PMOS switches off, so $V_{OL} = 0$ and the static power is zero.

PART 2 · LECTURE 24

Switching threshold, sizing and the symmetric inverter

One square root gives V_{th}. Run it backwards to size the transistors.

> **Big idea: V_{th} is where the tug-of-war is a tie, so it moves towards the weaker side's threshold.**

- ✓ Derive V_{th} (and pick the right sign)
- ✓ Why the PMOS is about 2.5× wider
- ✓ Read a CMOS layout
- ✓ Size $(W/L)_p$ for any target V_{th}
- ✓ The symmetric $3/8$ and $5/8$ results
- ✓ Use the inverter as an amplifier (Razavi)

2.1 Deriving V_{th} (point C: both saturated)

1. **Equate the currents** , both saturated:

$$\frac{k_n}{2}(V_{GS,n}-V_{T0,n})^2=\frac{k_p}{2}(V_{GS,p}-V_{T0,p})^2 \;\Rightarrow\; k_R(V_{in}-V_{T0,n})^2=(V_{in}-V_{DD}-V_{T0,p})^2$$

2. **Put $V_{in}=V_{th}$ and take square roots:** $\sqrt{k_R}\,(V_{th}-V_{T0,n})=\pm(V_{th}-V_{DD}-V_{T0,p})$.

   Which sign? The left side is positive (NMOS on). The bracket on the right is $V_{GS,p}-V_{T0,p}$, which is **negative** (PMOS on). So take the **minus** sign: $\sqrt{k_R}(V_{th}-V_{T0,n})=V_{DD}+V_{T0,p}-V_{th}$.

3. **Collect the V_{th} terms** and divide top and bottom by $\sqrt{k_R}$:

$$\boxed{V_{th}=\frac{V_{T0,n}+\sqrt{1/k_R}\,(V_{DD}+V_{T0,p})}{1+\sqrt{1/k_R}}}$$

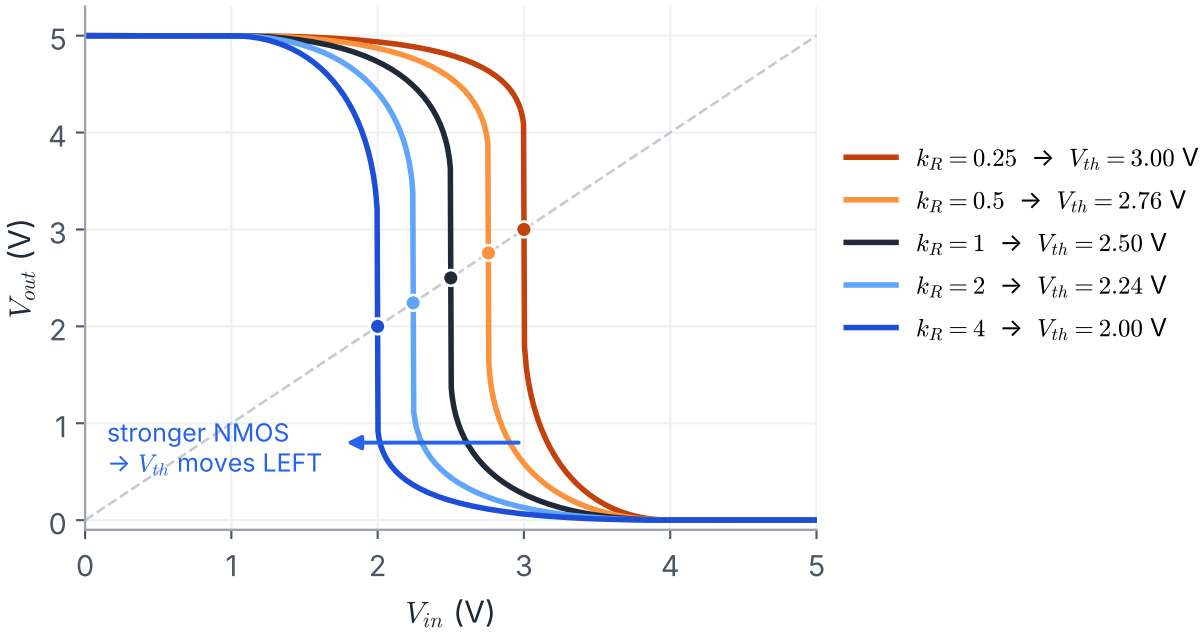


Fig. 2.1 VTCs for five values of k_R ($V_{DD} = 5$ V, $V_T = \pm 1$ V); dots mark V_{th}. A stronger NMOS ($k_R > 1$) wins the tug-of-war earlier, so the curve shifts **left**; a stronger PMOS shifts it **right**.

READ THE FORMULA AS A WEIGHTED AVERAGE

V_{th} always lies between $V_{T0,n}$ (where the NMOS starts) and $V_{DD} + V_{T0,p}$ (where the PMOS stops). The weight on the PMOS end is $\sqrt{1/k_R}$. As $k_R \to \infty$, $V_{th} \to V_{T0,n}$; as $k_R \to 0$, $V_{th} \to V_{DD} + V_{T0,p}$.

2.2 Design: run it backwards

Go back to step 2 and solve for k_R instead:

$$\sqrt{k_R} = \frac{V_{DD} + V_{T0,p} - V_{th}}{V_{th} - V_{T0,n}} \xrightarrow{V_{th} = V_{DD}/2} \boxed{k_R = \left(\frac{0.5V_{DD} + V_{T0,p}}{0.5V_{DD} - V_{T0,n}}\right)^2}$$

If $V_{T0,n} = |V_{T0,p}|$ the bracket is 1, so $k_R = 1$. Now expand k_R (same t_{ox}, so C_{ox} cancels):

$$\frac{k_n}{k_p} = \frac{\mu_n (W/L)_n}{\mu_p (W/L)_p} = 1 \;\Rightarrow\; \frac{(W/L)_p}{(W/L)_n} = \frac{\mu_n}{\mu_p} = \frac{580}{230} \;\Rightarrow\; \boxed{(W/L)_p \approx 2.5\,(W/L)_n}$$

SYMMETRIC INVERTER

$V_{T0,n} = |V_{T0,p}|$ and $k_R = 1$ give $V_{th} = V_{DD}/2$.
The PMOS is about **2.5× wider** to make up for the lower hole mobility.

2.3 The symmetric inverter's V_{IL} and V_{IH}

Put $k_R = 1$ and $V_{T0,p} = -V_{T0,n} = -V_T$ into ②. The thresholds cancel:

$$V_{IL} = \frac{2V_{out} - V_T - V_{DD} + V_T}{2} = V_{out} - 0.5V_{DD} \quad \Rightarrow \quad V_{out} - V_{DD} = V_{IL} - 0.5V_{DD}$$

The lecture then says "substitute into ①" and jumps to the answer. Here is the algebra in between, with $x = V_{IL}$:

1. ① becomes $(x - V_T)^2 = 2(x - V_{DD} + V_T)(x - \frac{V_{DD}}{2}) - (x - \frac{V_{DD}}{2})^2$.
2. Expand both sides; the x^2 terms cancel:
$$x^2 - 2V_T x + V_T^2 = x^2 - 2V_{DD}x + 2V_T x + \tfrac{3}{4}V_{DD}^2 - V_T V_{DD}$$
3. Collect: $x(2V_{DD} - 4V_T) = \frac{3}{4}V_{DD}^2 - V_T V_{DD} - V_T^2 = \frac{1}{4}(3V_{DD} + 2V_T)(V_{DD} - 2V_T)$.
4. Cancel $(V_{DD} - 2V_T)$: $\boxed{V_{IL} = \frac{1}{8}(3V_{DD} + 2V_T)}$. The same steps on ③ and ④ give $\boxed{V_{IH} = \frac{1}{8}(5V_{DD} - 2V_T)}$.

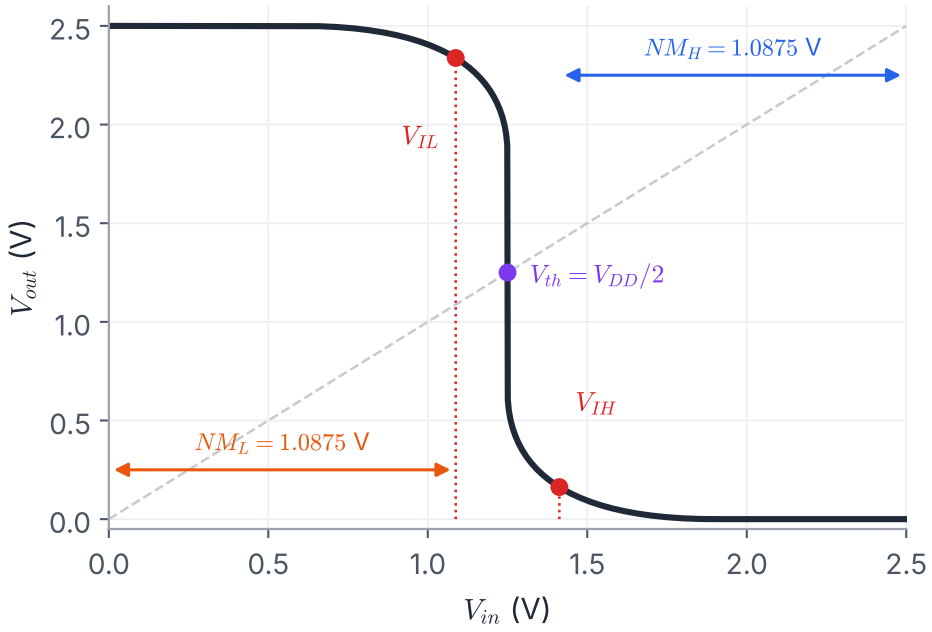


Fig. 2.2 A symmetric inverter ($V_{DD} = 2.5$ V, $V_T = 0.6$ V). The VTC is point-symmetric about $(V_{DD}/2, V_{DD}/2)$, so the two noise margins are equal. That is the reason to build it this way.

SYMMETRIC RESULTS

$$V_{IL} = \frac{3V_{DD} + 2V_T}{8}, \qquad V_{IH} = \frac{5V_{DD} - 2V_T}{8}$$

$$V_{IL} + V_{IH} = V_{DD}, \qquad NM_L = NM_H = V_{IL}$$

REMEMBER "3 + 5 = 8"

Low gets **3**, high gets **5**, both over **8**. The $\pm 2V_T$ signs are whatever makes the pair add to V_{DD}.

Why the margins are equal: $NM_H = V_{DD} - V_{IH} = V_{DD} - (V_{DD} - V_{IL}) = V_{IL} = NM_L$.

2.4 Layout

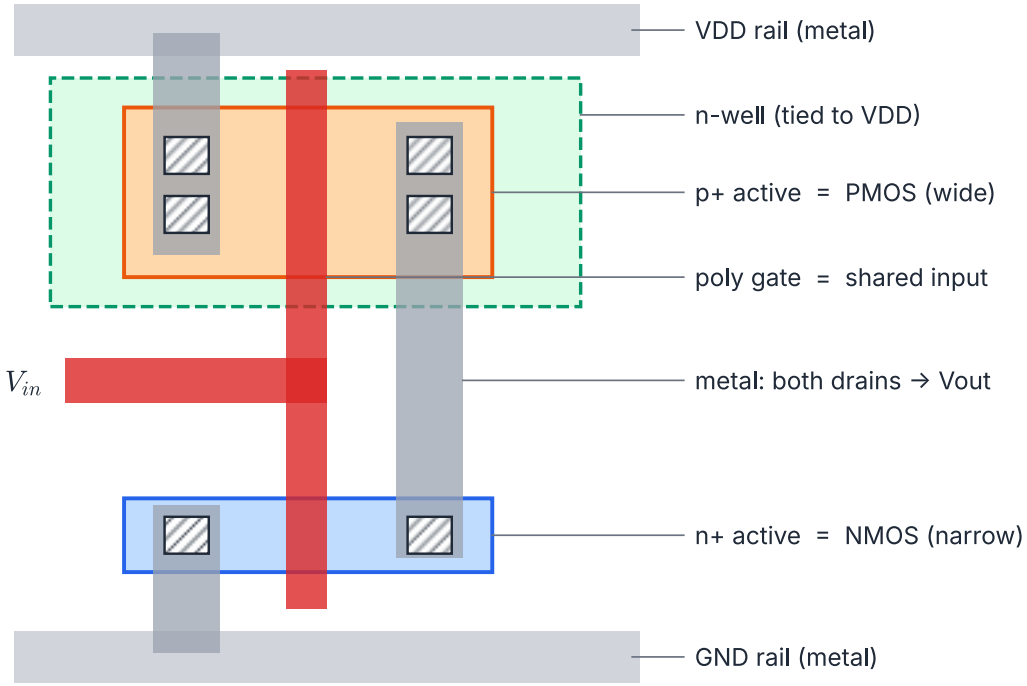


Fig. 2.3 CMOS inverter layout, top view (after the lecture sketch). One vertical poly line is the shared gate (V_{in}). Where it crosses the p⁺ region (inside the n-well), it makes the PMOS. Where it crosses the n⁺ region, it makes the NMOS. The sources connect to the V_{DD} and GND rails. A metal strap joins the two drains to make V_{out}. The p⁺ region is wider: this is the sizing of §2.2.

2.5 The inverter as an amplifier (Razavi §3.3.4)

Bias the inverter exactly at V_{th}: both transistors are saturated. A small rise in V_{in} makes the NMOS pull **more** current down *and* the PMOS push **less** current up. Both effects push V_{out} down, so they **add**. Razavi calls this a "CS stage with active load" (complementary CS stage):

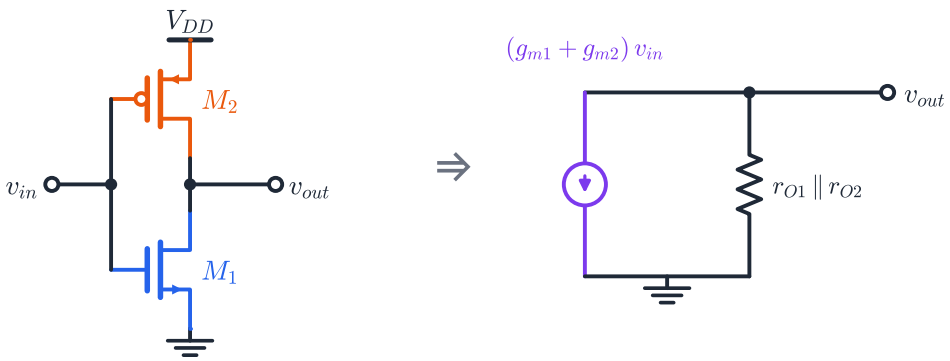


Fig. 2.4 For small signals the two devices are in parallel: transconductances add, and so do the output conductances (Razavi Fig. 3.20).

RAZAVI EQ. (3.48)

$$A_v = -(g_{m1} + g_{m2})\,(r_{O1} \parallel r_{O2}), \qquad g_m = k\,|V_{ov}|, \qquad r_O = \frac{1}{\lambda I_D}$$

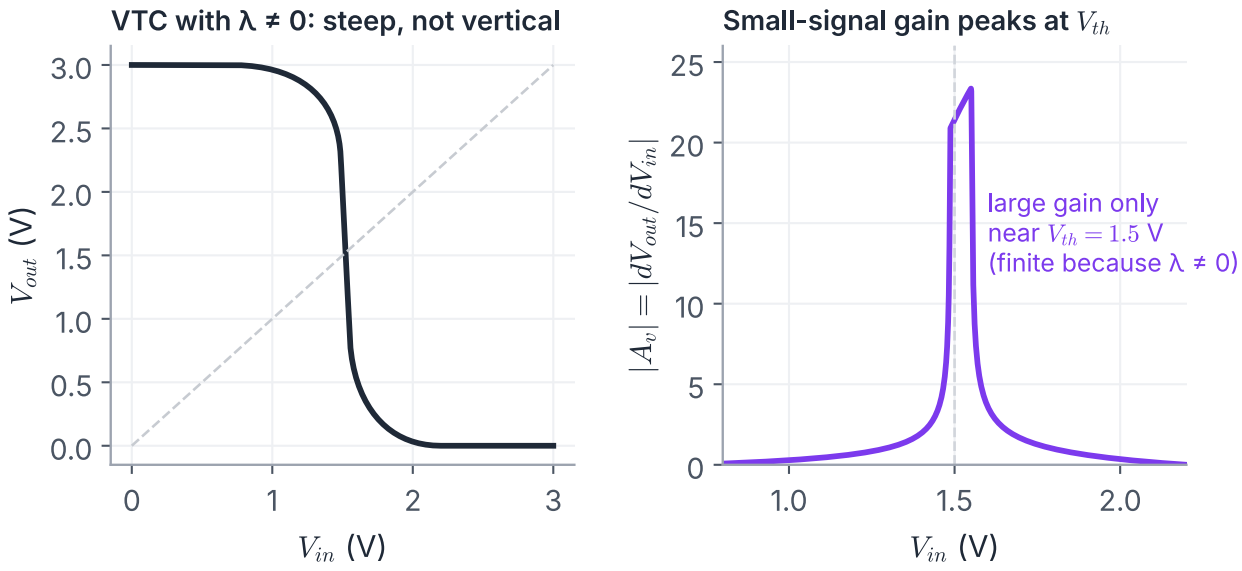
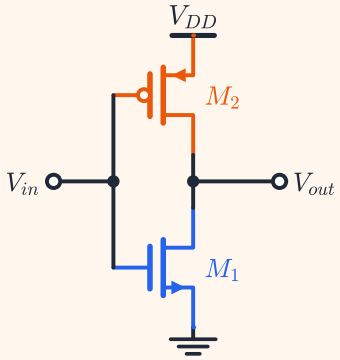


Fig. 2.5 With channel-length modulation ($\lambda \neq 0$), the VTC at C is steep but not vertical. The gain is large only in a narrow window around V_{th}. With $\lambda = 0$ the gain would be infinite, which is why the ideal VTC drops vertically.

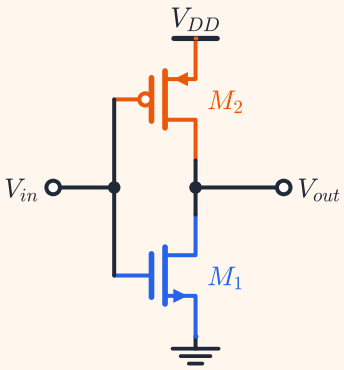
YOUR TURN · PART 2 (SOLUTIONS ON THE NEXT PAGE)

2.1 Tutorial 5 Q2 (2024–25) Find V_{th} and the noise margins.



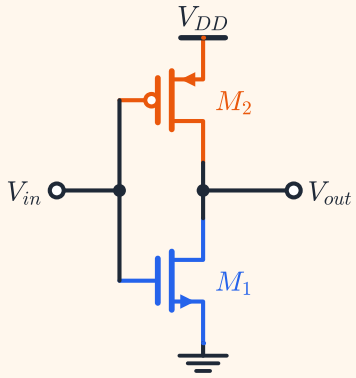
$V_{DD} = 3.3$ V
NMOS: 0.6 V, 60 µA/V², $W/L = 8$
PMOS: -0.7 V, 25 µA/V², $W/L = 12$
Find V_{th} and both NMs

2.2 Compre 2024–25 Q4(b) The inverter is sized for **equal rise and fall delays**. Find the noise margins. (Part 3 shows why equal delays mean $k_n = k_p$.)



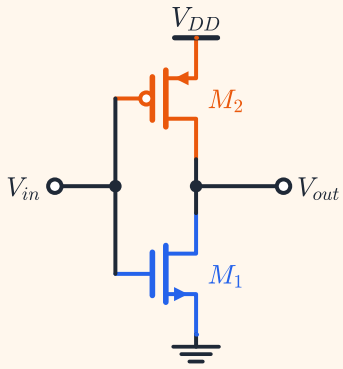
$V_{DD} = 2.5$ V, $V_{Tn} = -V_{Tp} = 0.6$ V
$k'_n = 3k'_p = 120$ µA/V²
sized for equal rise & fall
Find NM_L, NM_H

2.3 Find $\beta = (W/L)_p/(W/L)_n$ for $V_{th} = V_{DD}/2$.



$V_{DD} = 3.3$ V
$V_{T0,n} = 0.6$ V, $V_{T0,p} = -0.7$ V
$\mu_n/\mu_p = 580/230$, same t_{ox}
Want $V_{th} = V_{DD}/2$: β = ?

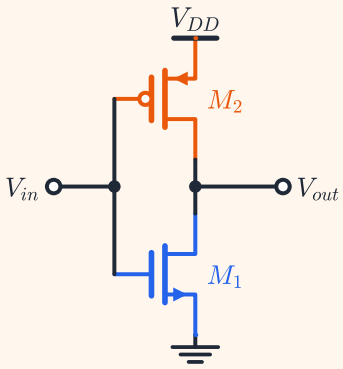
2.4 Find the new V_{th}. Do the margins stay equal, or does one grow at the other's expense?



$V_{DD} = 5$ V, $V_T = \pm 1$ V
symmetric design, then
redesigned with $k_R = 4$
New V_{th}? Margins?

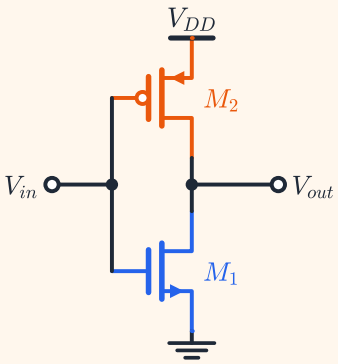
2.5 Prove $V_{IL} + V_{IH} = V_{DD}$ for the symmetric inverter *without* the closed forms, using the 180° symmetry of the VTC.

2.6 Razavi §3.3.4, Eq. (3.48), Table 2.1 (a) Choose $(W/L)_p$ so that $V_{th} = 1.5$ V. (b) Find the small-signal gain at V_{th} using $A_v = -(g_{m1} + g_{mb2})(r_{O1} \| r_{O2})$.



Razavi Table 2.1 devices, $V_{DD} = 3$ V
$V_{THN} = 0.7$, $|V_{THP}| = 0.8$ V
$\lambda_n = 0.1$, $\lambda_p = 0.2$ V^{-1}
$(W/L)_n = 10/0.5$, want $V_{th} = 1.5$ V

2.7 Razavi Table 2.1 Both devices drawn the same size. Find V_{th} and say which way it moved from $V_{DD}/2$ and why.



Razavi Table 2.1 devices, $V_{DD} = 3$ V
$\mu_n C_{ox} = 134.3$, $\mu_p C_{ox} = 38.4$ µA/V²
$(W/L)_n = (W/L)_p = 10/0.5$
Find V_{th}

Solutions · Part 2

2.1 $k_n = 60 \times 8 = 480$, $k_p = 25 \times 12 = 300$ μA/V², so $k_R = 1.6$ and $\sqrt{1/k_R} = 0.7906$.

$$V_{th} = \frac{0.6 + 0.7906 \times 2.6}{1.7906} = \mathbf{1.483\ V}$$

V_{IL}: ② gives $V_{out} = \frac{1}{2}(2.6V_{IL} + 3.04)$; into ①: $V_{IL} = 1.199$ **V** $= NM_L$. V_{IH}: $V_{IH} = 1.696$ **V, so** $NM_H = 1.604$ **V**.
The printed key gives 1.48, 1.16, 1.69 and 1.61 V. A brute-force simulation of the VTC confirms 1.199 V for V_{IL}, so the key's 1.16 is slightly off.

2.2 Equal delays need equal strengths, $k_n = k_p$, so $(W/L)_p = 3(W/L)_n$ and $k_R = 1$ with matched thresholds: the inverter is symmetric. $V_{IL} = (7.5 + 1.2)/8 = 1.0875$ V, $V_{IH} = (12.5 - 1.2)/8 = 1.4125$ V, so $NM_L = NM_H = 1.0875$ **V** and $V_{th} = 1.25$ V.

2.3

$$k_R = \left(\frac{1.65 - 0.7}{1.65 - 0.6}\right)^2 = 0.8186, \qquad \beta = \frac{(W/L)_p}{(W/L)_n} = \frac{\mu_n}{\mu_p\, k_R} = \frac{2.522}{0.8186} = \mathbf{3.08}$$

The larger $|V_{T0,p}|$ weakens the PMOS further, so it must be even wider than 2.5×.

2.4 $V_{th} = \dfrac{1 + 0.5 \times 4}{1.5} = \mathbf{2.0\ V}$ (moved left). Shortcut: $V_{IL} = 1.512$ V and $V_{IH} = 2.219$ V, so $NM_L = 1.51$ **V,** $NM_H = 2.78$ **V**. No longer equal: NM_H grew while NM_L shrank from 1.75 V. The symmetric design maximises the *worse* of the two margins.

2.5 In the symmetric case the PMOS is the NMOS with every voltage measured from the other rail. So if (V_{in}, V_{out}) lies on the VTC, so does $(V_{DD} - V_{in},\ V_{DD} - V_{out})$: a 180° rotation about $(V_{DD}/2, V_{DD}/2)$. A rotation by 180° keeps slopes unchanged, so it maps the slope −1 point V_{IL} to the other slope −1 point V_{IH}. Therefore $V_{IH} = V_{DD} - V_{IL}$.

2.6 (a) From §2.2: $k_R = \left(\dfrac{3 - 0.8 - 1.5}{1.5 - 0.7}\right)^2 = 0.7656$. Then $k_n = 134.3\mu \times 20 = 2.686$ mA/V² and $k_p = k_n/k_R = 3.508$ mA/V², so

$$(W/L)_p = \frac{3.508\text{m}}{38.4\mu} = 91.4 \ \Rightarrow\ \textbf{about}\ \ 45.7/0.5$$

(b) $I_D = \frac{k_n}{2}(0.8)^2 = 0.860$ mA. $g_{m1} = k_n(0.8) = 2.149$ mS, $g_{m2} = k_p(0.7) = 2.456$ mS. $r_{O1} = 1/(0.1I_D) = 11.63$ kΩ, $r_{O2} = 1/(0.2I_D) = 5.82$ kΩ, parallel $= 3.88$ kΩ.

$$A_v = -(4.605\ \text{mS})(3.88\ \text{k}\Omega) \approx \mathbf{-17.9}$$

This finite gain is why a real VTC is steep but not vertical at C.

2.7 Same W/L, so $k_R = \mu_n C_{ox}/\mu_p C_{ox} = 134.3/38.4 = 3.50$ and $\sqrt{1/k_R} = 0.535$.

$$V_{th} = \frac{0.7 + 0.535 \times (3 - 0.8)}{1.535} = \mathbf{1.22\ V}$$

It moved **left** of 1.5 V: the same-size NMOS is much stronger, so it wins the tug-of-war early.

PART 3 · LECTURE 25

Switching characteristics and delay

The output is a capacitor. The transistors are the currents that charge and discharge it.

> **Big idea: delay = (charge to move) ÷ (current available), i.e. $\tau \approx C\,\Delta V / I$.**

- ✓ List every capacitor that loads V_{out}
- ✓ Define τ_{PHL}, τ_{PLH}, τ_P, rise and fall
- ✓ Average-current method (quick)
- ✓ Exact method by integration
- ✓ Size for equal rise and fall
- ✓ Ring oscillators and dynamic power

3.1 Where the load capacitance comes from

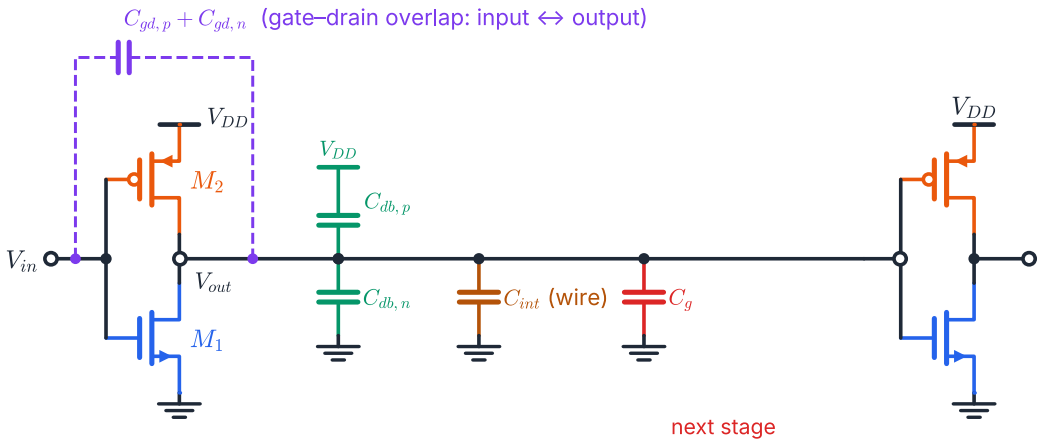


Fig. 3.1 An inverter driving a second inverter. Every capacitor with one plate on V_{out} is part of the load. These are the gate–drain overlaps (purple), the drain junctions (green), the wire (amber) and the next stage's gate (red). C_{gs} sits between the gate and a fixed rail, so it loads the *input*, not the output.

LUMPED LOAD

$$C_{load} = C_{gd,p} + C_{gd,n} + C_{db,p} + C_{db,n} + C_{int} + C_g, \qquad C_g \approx C_{ox}\left[(WL)_n + (WL)_p\right]$$

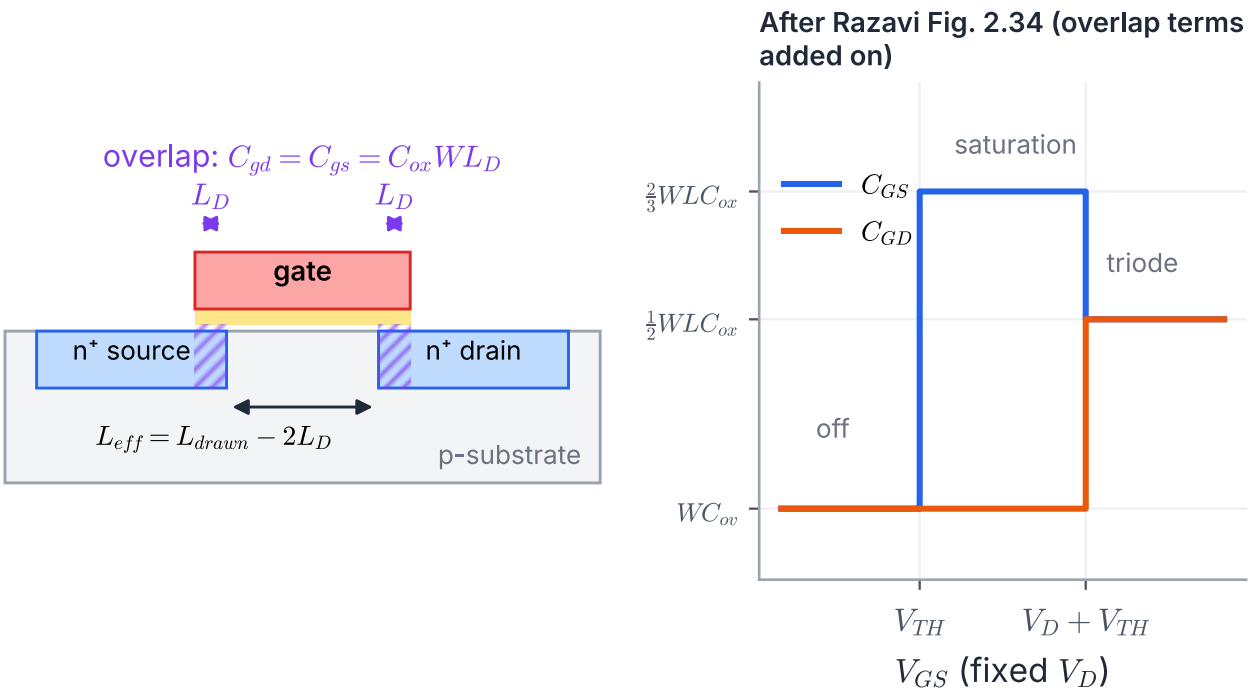


Fig. 3.2 Left: the source and drain diffuse a distance L_D under the gate; that overlap gives $C_{gd} = C_{gs} = C_{ox}WL_D$ ($= WC_{ov}$ in Razavi). Right: Razavi's bias dependence. In saturation the channel connects to the source, so C_{GD} is overlap only.

From Razavi §2.4.2: the junction capacitance is $C_{db} = C_j \cdot (\text{drain area}) + C_{jsw} \cdot (\text{drain perimeter})$. For a drain of width W and length E: $C_{db} = WEC_j + 2(W + E)C_{jsw}$. A folded layout (Razavi Ex. 2.10) halves the drain area and so shrinks C_{db}.

3.2 The lumped model

All the physics becomes one node equation (KCL at V_{out}). The current into the capacitor is the PMOS current in, minus the NMOS current out.

$$i_C = i_{D,p} - i_{D,n} = C_{load}\frac{dV_{out}}{dt}$$



Fig. 3.3 Falling output (left): the input jumps high, the PMOS turns off, and the NMOS drains the capacitor to ground. Rising output (right): the input drops, the NMOS turns off, and the PMOS fills the capacitor from V_{DD}. Only one transistor works in each case.

3.3 Definitions

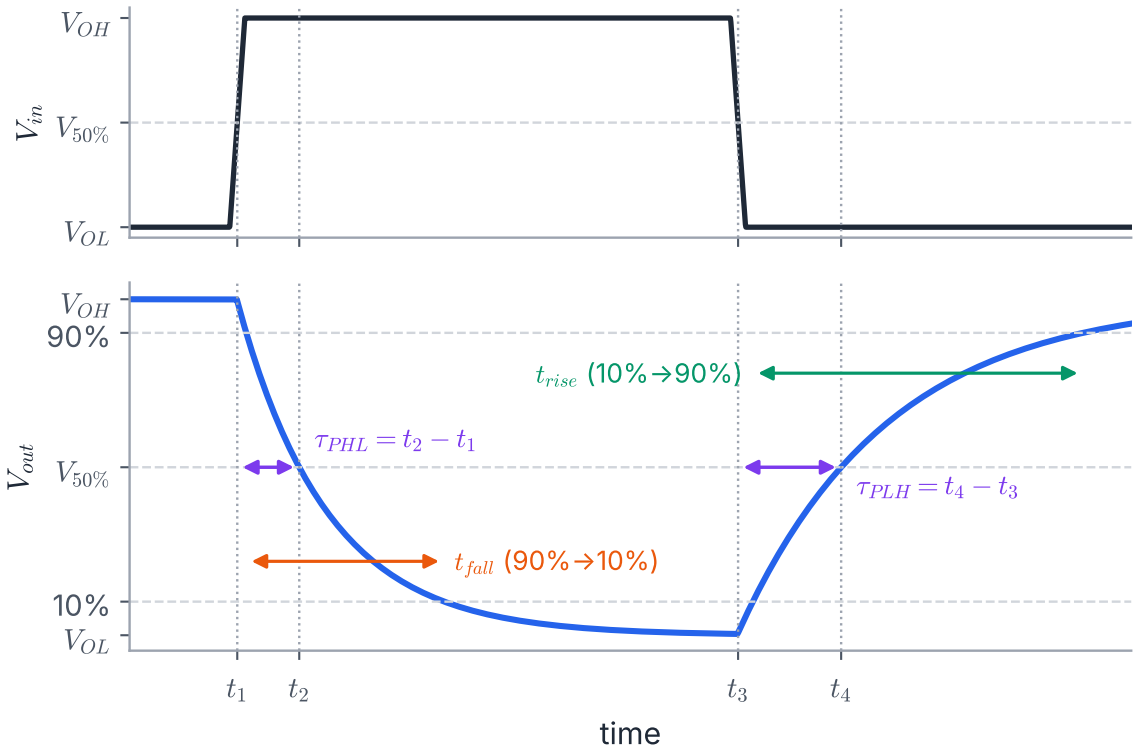


Fig. 3.4 Delays are measured between the 50 % points of input and output. Rise and fall times are measured on the output alone, between 10 % and 90 %.

QUANTITY	DEFINITION	REMEMBER
$V_{50\%}$	$V_{OL} + \frac{1}{2}(V_{OH} - V_{OL}) = \frac{V_{OH} + V_{OL}}{2}$	halfway
τ_{PHL}	$t_2 - t_1$	output goes High → Low
τ_{PLH}	$t_4 - t_3$	output goes Low → High
τ_P	$\frac{1}{2}(\tau_{PHL} + \tau_{PLH})$	average delay
t_{rise}, t_{fall}	$t_{90\%} - t_{10\%}$ (rise), $t_{10\%} - t_{90\%}$ (fall)	output only

3.4 Method 1: average current

Pretend the capacitor current is constant at its average value. Then $i = C\,dV/dt$ becomes simply $\tau = C\,\Delta V / I_{avg}$:

$$\tau_{PHL} = \frac{C_{load}\,\Delta V_{HL}}{I_{avg,HL}}, \qquad \Delta V_{HL} = V_{OH} - V_{50\%}, \qquad I_{avg,HL} = \tfrac{1}{2}\left[i_{D,n}\big|_{V_{out}=V_{OH}} + i_{D,n}\big|_{V_{out}=V_{50\%}}\right] \;\; (V_{in} = V_{OH})$$

Likewise $\tau_{PLH} = C_{load}\Delta V_{LH}/I_{avg,LH}$ with $\Delta V_{LH} = V_{50\%} - V_{OL}$ and the PMOS current averaged at $V_{out} = V_{OL}$ and $V_{out} = V_{50\%}$ (with $V_{in} = V_{OL}$).

PICK THE RIGHT REGION AT EACH ENDPOINT

At the start of the fall ($V_{out} = V_{OH}$): $V_{DS,n} = V_{OH} \geq V_{OH} - V_{T0,n}$, so the NMOS is **saturated**. At $V_{50\%}$: usually $V_{50\%} < V_{OH} - V_{T0,n}$, so it is **linear**. Using the saturation formula at both ends is the most common mistake.

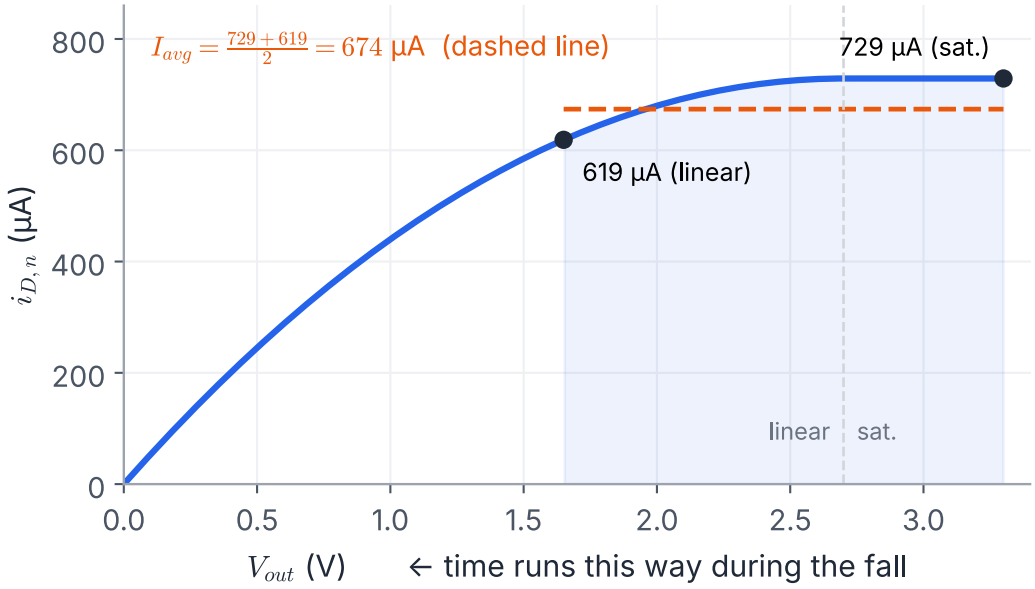


Fig. 3.5 The NMOS current during the fall (Worked example 25.2 numbers). The method replaces the real curve over the shaded range by the mean of its two end values.

3.5 Method 2: exact integration

Step the input from V_{OL} to V_{OH} at t_1. As V_{out} falls, the NMOS passes through **two regions**, so the integral splits in two:

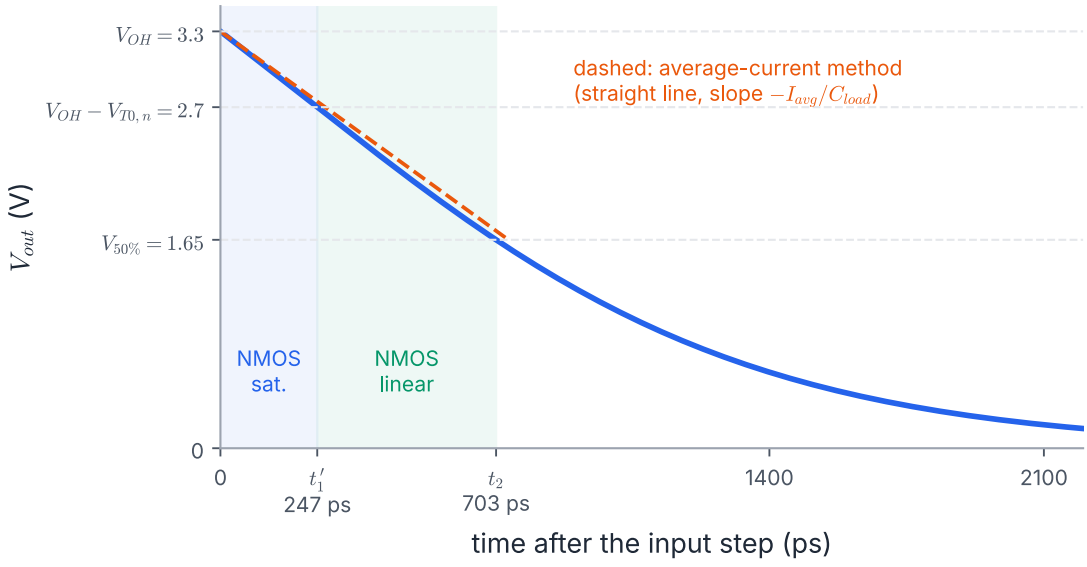


Fig. 3.6 The exact $V_{out}(t)$. From t_1 to t_1' the NMOS is saturated: constant current, so a straight line. From t_1' to t_2 it is linear and the discharge curves (current falls as V_{out} falls).

Segment 1 · saturated, $V_{OH} \to V_{OH} - V_{T0,n}$

The current is the constant $\frac{k_n}{2}(V_{OH} - V_{T0,n})^2$, so time = charge ÷ current:

$$t_1' - t_1 = \frac{C_{load}\, V_{T0,n}}{\frac{k_n}{2}(V_{OH} - V_{T0,n})^2} = \boxed{\frac{2C_{load} V_{T0,n}}{k_n (V_{OH} - V_{T0,n})^2}}$$

Segment 2 · linear, $V_{OH} - V_{T0,n} \to V_{50\%}$

Now $i_{D,n} = \frac{k_n}{2}\left[2(V_{OH} - V_{T0,n})V_{out} - V_{out}^2\right]$. Write $a = 2(V_{OH} - V_{T0,n})$ so the bracket is $V_{out}(a - V_{out})$, and use the integral of §0.3:

$$t_2 - t_1' = -\frac{2C_{load}}{k_n}\int_{a/2}^{V_{50\%}} \frac{dV}{V(a-V)} = \frac{2C_{load}}{k_n\, a}\left[\ln\frac{V}{a-V}\right]_{V_{50\%}}^{a/2}$$

At the upper limit $V = a/2$, so $\ln\frac{V}{a-V} = \ln 1 = 0$. That leaves

$$t_2 - t_1' = \frac{C_{load}}{k_n(V_{OH} - V_{T0,n})}\ln\left(\frac{2(V_{OH} - V_{T0,n})}{V_{50\%}} - 1\right)$$

MEMORISE: EXACT STEP-INPUT DELAYS

$$\tau_{PHL} = \frac{C_{load}}{k_n(V_{OH} - V_{T0,n})}\left[\frac{2V_{T0,n}}{V_{OH} - V_{T0,n}} + \ln\left(\frac{4(V_{OH} - V_{T0,n})}{V_{OH} + V_{OL}} - 1\right)\right]$$

$$\tau_{PLH} = \frac{C_{load}}{k_p(V_{OH} - V_{OL} - |V_{T0,p}|)}\left[\frac{2|V_{T0,p}|}{V_{OH} - V_{OL} - |V_{T0,p}|} + \ln\left(\frac{4(V_{OH} - V_{OL} - |V_{T0,p}|)}{V_{OH} - V_{OL}} - 1\right)\right]$$

For CMOS ($V_{OH} = V_{DD}$, $V_{OL} = 0$) both are $\frac{C}{k(V_{DD} - |V_T|)}\left[\frac{2|V_T|}{V_{DD} - |V_T|} + \ln\left(\frac{4(V_{DD} - |V_T|)}{V_{DD}} - 1\right)\right]$ with the right transistor's k and V_T.

SAY IT OUT LOUD

"**C over kV, times [sat bit + log bit]**." The sat bit is $2V_T/(V_{DD} - V_T)$. The log bit is $\ln(4 \cdot \text{overdrive}/V_{DD} - 1)$.

WORKED EXAMPLE 25.2

$V_{DD} = 3.3$ V, $C_{load} = 0.3$ pF, $k_n = 200$, $k_p = 80$ μA/V², $V_{T0,n} = 0.6$ V, $V_{T0,p} = -0.7$ V.

Average current, HL. At $V_{out} = 3.3$ (sat.): $100\mu(2.7)^2 = 729$ μA. At $V_{out} = 1.65$ (linear): $100\mu[2(2.7)(1.65) - 1.65^2] = 619$ μA. Average 674 μA, so $\tau_{PHL} = 0.3\text{p} \times 1.65/674\mu = \mathbf{0.734}$ ns.

Exact, HL. $\dfrac{0.3\text{p}}{200\mu \times 2.7} = 0.556$ ns, times $\left[\frac{1.2}{2.7} + \ln\left(\frac{10.8}{3.3} - 1\right)\right] = 0.444 + 0.821$, gives $\tau_{PHL} = 0.247 + 0.456 = \mathbf{0.703}$ ns.

Exact, LH. $\dfrac{0.3\text{p}}{80\mu \times 2.6}\left[\frac{1.4}{2.6} + \ln\left(\frac{10.4}{3.3} - 1\right)\right] = 1.442 \times (0.538 + 0.766)$, so $\tau_{PLH} = \mathbf{1.88}$ ns (average-current method: 1.96 ns). Then $\tau_P = \mathbf{1.29}$ ns.

The average-current method overestimates by about 4 %. That is good enough for most questions, but use whichever method the question names.

3.6 What the formulas tell a designer

Both delays have the form $\tau \propto \dfrac{C_{load}}{k\,(V_{DD} - |V_T|)}$. Razavi's one-line estimate (§17.1) is the same idea: $T_d \approx C\,V_{DD}/I$. To go faster:

- **Reduce C_{load}:** shorter wires, smaller fan-out, folded drains.
- **Increase k:** make W/L larger. But wider devices add their own C_{db} and load the previous stage, so the gain is limited.
- **Increase V_{DD}:** faster, but dynamic power $\propto CV_{DD}^2 f$ rises (§3.8).

Equal rise and fall need $k_p = k_n$ (with matched thresholds): the same $(W/L)_p \approx 2.5(W/L)_n$ as Lecture 24's symmetric inverter. That is why "sized for equal delays" in Compre Q4 means "symmetric".

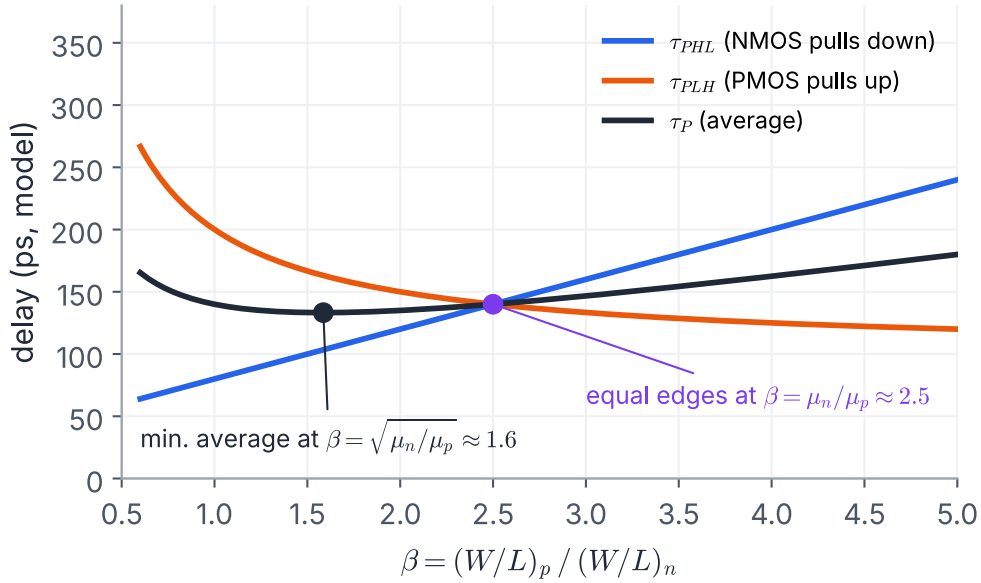


Fig. 3.7 Delay against PMOS/NMOS sizing when the load is mostly the inverter's own and the next stage's capacitance (simple model). A wider PMOS speeds up τ_{PLH} but adds capacitance, so τ_{PHL} rises. Equal edges happen at $\beta = \mu_n/\mu_p \approx 2.5$; the *average* is smallest a little earlier, at $\beta = \sqrt{\mu_n/\mu_p} \approx 1.6$. Equal edges and minimum average delay are slightly different goals.

3.7 Ring oscillator (Razavi §15.2)

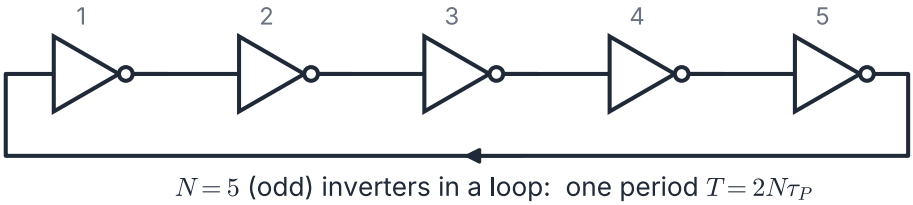


Fig. 3.8 An odd number of inverters in a loop cannot settle: each stage flips the next. An edge must travel round the loop **twice** (once as a rising edge, once as a falling edge) for one full period.

RING OSCILLATOR

$$T = 2N\tau_P \qquad f = \frac{1}{2N\tau_P} \qquad N \text{ odd}$$

Razavi: three stages give a period of $6T_D$, five stages give $f = 1/(10T_D)$.

3.8 Dynamic power

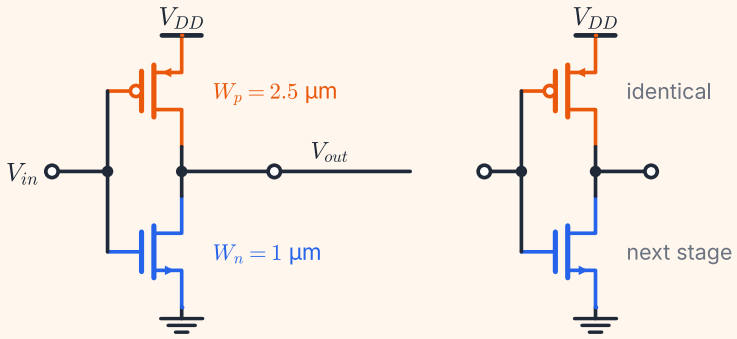
Each time the output rises, the PMOS takes charge $Q = C_{load}V_{DD}$ from the supply. That costs energy $QV_{DD} = C_{load}V_{DD}^2$. Half is stored in the capacitor. Half is burned in the PMOS. When the output falls, the NMOS burns the stored half. So each full $0 \to 1 \to 0$ cycle costs $C_{load}V_{DD}^2$:

MEMORISE

$$P_{dyn} = C_{load}\, V_{DD}^2\, f \qquad \text{(one full cycle per period; static power} \approx 0 \text{ for CMOS)}$$

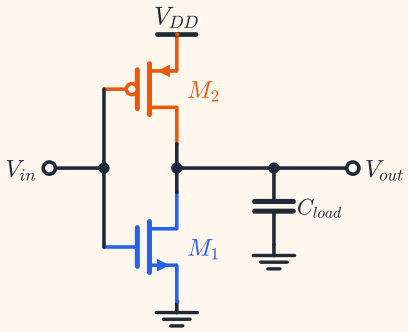
YOUR TURN · PART 3 (SOLUTIONS ON THE NEXT PAGES)

3.1 Take $C_{ox} = 3.8$ fF/µm², $L_D = 0.08$ µm, $L = 0.5$ µm, $W_n = 1$ µm, $W_p = 2.5$ µm, $C_{db,n} = 2$ fF, $C_{db,p} = 4$ fF, $C_{int} = 5$ fF. Find C_{load}. Which term dominates?



3.2 Derive segment 2 of §3.5 yourself, starting from $C\, dV_{out}/dt = -i_{D,n}$ in the linear region. Check that you reach the ln term.

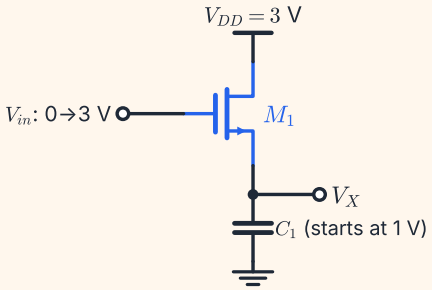
3.3 What $(W/L)_p$ multiplier makes $\tau_{PLH} = \tau_{PHL}$? Assume C_{load} does not change.



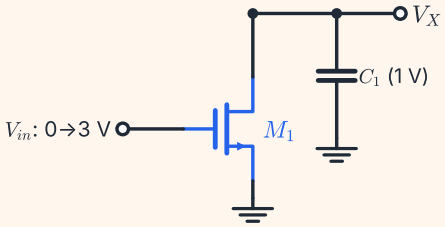
Worked example 25.2 inverter
$V_{DD} = 3.3$ V, $k_n = 200$, $k_p = 80$ µA/V²
$C_{load} = 0.3$ pF (fixed)
Scale $(W/L)_p$ so $\tau_{PLH} = \tau_{PHL}$

3.4 Razavi Prob. 2.11(b), numbers added An NMOS has $W/L = 50/0.5$, $\mu_n C_{ox} = 134.3\ \mu$A/V², $V_{TH} = 0.7$ V and $\lambda = \gamma = 0$. Its source is grounded and its drain is node X. $C_1 = 1$ pF connects X to ground and starts at 1 V. At $t = 0$ the gate steps from 0 to 3 V. Sketch $V_X(t)$ and find when $V_X = 0.5$ V. (Figure (b) below.)

3.5 Razavi Prob. 2.11(a), numbers added Same device, but now $V_{DD} = 3$ V is on the drain and C_1 (initially 1 V) is on the source. The gate steps from 0 to 3 V. Sketch $V_X(t)$, give its final value, and find when $V_X = 2.0$ V.



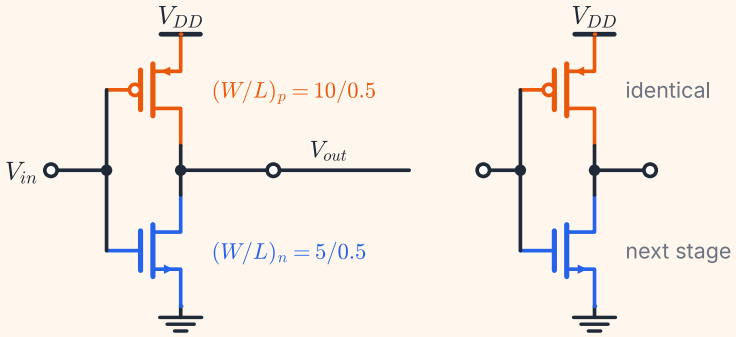
(a) Q3.5: source follower



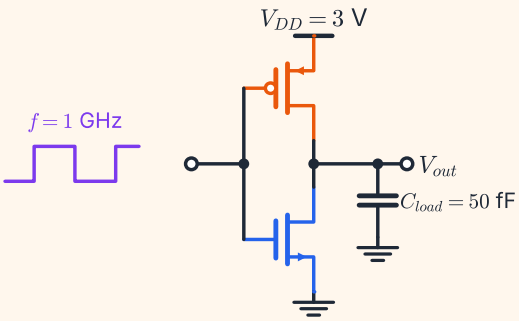
(b) Q3.4: discharge to ground

3.6 Razavi §15.2 A ring of 5 inverters, each with $\tau_P = 50$ ps, oscillates. What is its frequency? (Fig. 3.8.)

3.7 Razavi Prob. 2.15 style, Table 2.1 An inverter with $(W/L)_n = 5/0.5$, $(W/L)_p = 10/0.5$ drives an identical inverter. Use $C_{ox} = 3.84$ fF/μm², $C_{GDO} = 0.4$ (n) and 0.3 (p) fF/μm, $C_j = 0.56$ (n) and 0.94 (p) fF/μm², $C_{jsw} = 0.035$ (n) and 0.032 (p) fF/μm, drain length $E = 1.5$ μm, zero-bias junctions, no wire. Find C_{load}, then estimate the delay with $T_d \approx CV_{DD}/I$ using the NMOS saturation current ($V_{DD} = 3$ V).



3.8 Find the dynamic power, and the energy burned in the PMOS each cycle.



One full $0 \to 1 \to 0$ cycle per clock period
Find P_{dyn}

Solutions · Part 3

3.1 $C_{gd,n} = 3.8 \times 1 \times 0.08 = 0.30$ fF, $C_{gd,p} = 3.8 \times 2.5 \times 0.08 = 0.76$ fF, $C_g = 3.8 \times (1 + 2.5) \times 0.5 = 6.65$ fF.

$$C_{load} = 0.30 + 0.76 + 2 + 4 + 5 + 6.65 = \mathbf{18.7\ fF}$$

The next stage's gate dominates, closely followed by the wire. Some texts double C_{gd} for the Miller effect (both ends swing in opposite directions); the lecture counts it once.

3.2 See §3.5. The two keys are the partial fractions $\frac{1}{V(a-V)} = \frac{1}{a}\left(\frac{1}{V} + \frac{1}{a-V}\right)$ and noticing that the starting point is $V = a/2$, which kills one log.

3.3 With C_{load} fixed, $\tau_{PLH} \propto 1/k_p$. So scale k_p by $1.882/0.703 = \mathbf{2.68}$, i.e. $(W/L)_p$ about 2.7× its present value. (This is larger than $k_n/k_p = 2.5$ because $|V_{T0,p}| > V_{T0,n}$.)

3.4 $V_{GS} = 3$ V and $V_{DS} \le 1 < 2.3$ V. So the device is **linear the whole time**. V_X falls towards 0, almost like an exponential (Fig. 3.9b). This is segment 2 alone, with $V_{ov} = 2.3$ V and $k_n = 13.43$ mA/V²:

$$t = \frac{C}{k_n V_{ov}} \ln \frac{V_0(2V_{ov} - V)}{V(2V_{ov} - V_0)}, \qquad \frac{C}{k_n V_{ov}} = 32.4\ \text{ps}, \quad \ln \frac{1 \times 4.1}{0.5 \times 3.6} = 0.823$$

so $t \approx 26.6$ **ps**.

3.5 $V_{DS} = 3 - V_X \geq V_{GS} - V_{TH} = 2.3 - V_X$ always, so the device is **saturated** throughout: a source follower charging C_1.

$$C\frac{dV_X}{dt} = \frac{k_n}{2}(2.3 - V_X)^2 \;\Rightarrow\; \left[\frac{1}{2.3 - V}\right]_1^V = \frac{k_n t}{2C} \;\Rightarrow\; t = \frac{2C}{k_n}\left(\frac{1}{0.3} - \frac{1}{1.3}\right) = \mathbf{0.38\ ns}$$

V_X rises towards $V_{in} - V_{TH} = 2.3$ **V** but never quite reaches it, because the current dies as $V_{GS} \to V_{TH}$. (With $\gamma > 0$ the final value is lower, because V_{TH} rises with V_{SB}.)

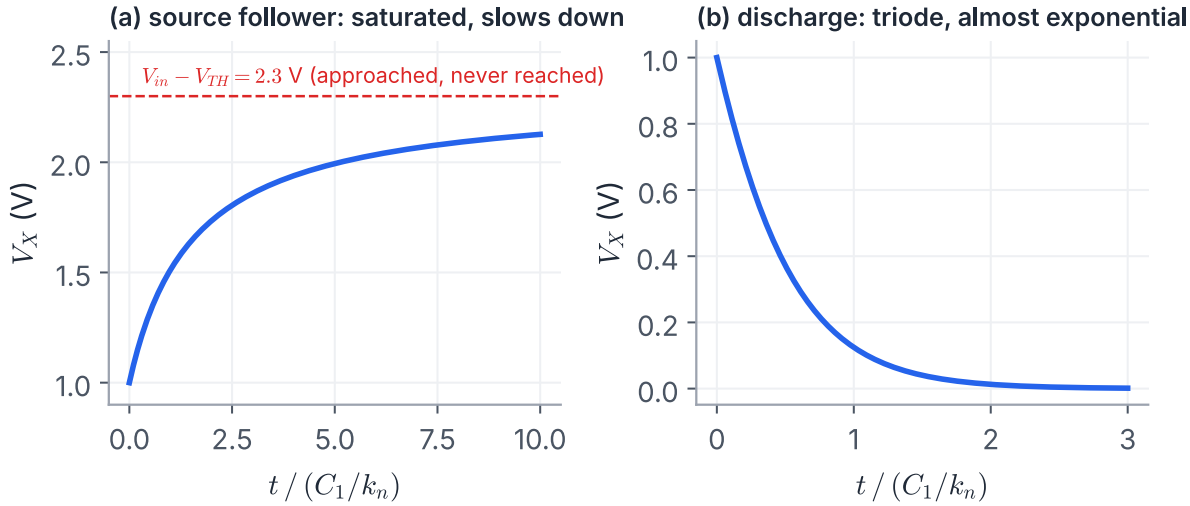


Fig. 3.9 The answers to Q3.5 (a) and Q3.4 (b), with time normalised to C_1/k_n. The follower slows down as it approaches its final value; the discharge is almost exponential.

3.6 $T = 2N\tau_P = 2 \times 5 \times 50 = 500$ ps, so $f = 2$ **GHz**. (The number of stages must be odd, or the ring latches instead of oscillating.)

3.7 $C_{gd,n} = 0.4 \times 5 = 2.0$ fF, $C_{gd,p} = 0.3 \times 10 = 3.0$ fF.
$C_{db,n} = 5(1.5)(0.56) + 2(5 + 1.5)(0.035) = 4.66$ fF, $C_{db,p} = 10(1.5)(0.94) + 2(10 + 1.5)(0.032) = 14.84$ fF.
$C_g = 3.84 \times (5 \times 0.5 + 10 \times 0.5) = 28.8$ fF. Total $C_{load} \approx 53.3$ **fF**.
$I = \frac{134.3\mu \times 10}{2}(3 - 0.7)^2 = 3.55$ mA, so $T_d \approx \dfrac{53.3\text{f} \times 3}{3.55\text{m}} \approx$ **45 ps**. Notice the PMOS drain junction is the second-largest term: wide devices are not free.

3.8 $P = CV_{DD}^2 f = 50\text{f} \times 9 \times 10^9 =$ **0.45 mW**. Energy per cycle $= CV_{DD}^2 = 0.45$ pJ, of which **0.225 pJ** is burned in the PMOS (while charging) and 0.225 pJ in the NMOS (while discharging).

PART 4 · REVISION

Formula sheet and revision

Everything from Lectures 23–25 that you should be able to write from memory.

RESULT	FORMULA	WHEN
Boundary lines	$V_{out} = V_{in} - V_{T0,n}$; $V_{out} = V_{in} - V_{T0,p}$	NMOS sat. above; PMOS sat. below
V_{OH}, V_{OL}	V_{DD}, 0	always (CMOS)
V_{IL} (with ①)	$\dfrac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R}$	n sat., p lin., slope −1
V_{IH} (with ③)	$\dfrac{V_{DD} + V_{T0,p} + k_R(2V_{out} + V_{T0,n})}{1 + k_R}$	n lin., p sat., slope −1
V_{th}	$\dfrac{V_{T0,n} + \sqrt{1/k_R}(V_{DD} + V_{T0,p})}{1 + \sqrt{1/k_R}}$	both sat.
k_R for a target V_{th}	$\left(\dfrac{V_{DD} + V_{T0,p} - V_{th}}{V_{th} - V_{T0,n}}\right)^2$	design
Symmetric sizing	$(W/L)_p = \dfrac{\mu_n}{\mu_p}(W/L)_n \approx 2.5\,(W/L)_n$	$V_{T0,n} = \lvert V_{T0,p}\rvert$, $k_R = 1$
Symmetric V_{IL}, V_{IH}	$\frac{1}{8}(3V_{DD} + 2V_T)$; $\frac{1}{8}(5V_{DD} - 2V_T)$; $NM_L = NM_H = V_{IL}$	$k_R = 1$
Noise margins	$NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$	
Amplifier gain	$A_v = -(g_{m1} + g_{m2})(r_{O1} \lVert r_{O2})$	biased at V_{th}
Load	$C_{gd,p} + C_{gd,n} + C_{db,p} + C_{db,n} + C_{int} + C_g$	$C_{gd} = C_{ox} W L_D$
$V_{50\%}$, τ_P	$\frac{1}{2}(V_{OH} + V_{OL})$; $\frac{1}{2}(\tau_{PHL} + \tau_{PLH})$	50 % to 50 %
Average current	$\tau = C_{load}\,\Delta V / I_{avg}$, $\Delta V = \frac{1}{2}(V_{OH} - V_{OL})$	average of the 2 endpoints
Saturated segment	$t_1' - t_1 = \dfrac{2C_{load}V_{T0,n}}{k_n(V_{OH} - V_{T0,n})^2}$	$V_{OH} \to V_{OH} - V_{T0,n}$
Exact τ_{PHL}	$\dfrac{C_{load}}{k_n(V_{OH} - V_{T0,n})}\left[\dfrac{2V_{T0,n}}{V_{OH} - V_{T0,n}} + \ln\left(\dfrac{4(V_{OH} - V_{T0,n})}{V_{OH} + V_{OL}} - 1\right)\right]$	step input

Exact τ_{PLH}	same with $n \to p$, $V_{OH} - V_{T0,n} \to V_{OH} - V_{OL} - \lvert V_{T0,p}\rvert$, $V_{OH} + V_{OL} \to V_{OH} - V_{OL}$	step input
Ring, power	$f = 1/(2N\tau_P)$; $P = C_{load} V_{DD}^2 f$	N odd

Memory tricks

Regions A–E

"Off-Lin, Sat-Lin, Sat-Sat, Lin-Sat, Lin-Off" (NMOS first). Weakly on = saturated; strongly on = linear.

Pictures

Off = open switch. Linear = resistor. Saturated = current source.

V_{IL} vs V_{IH}

Both: $1 + k_R$ below, $2V_{out}$ on top. V_{IL}: k_R only on $V_{T0,n}$. V_{IH}: k_R on $(2V_{out} + V_{T0,n})$.

V_{th} as a tug-of-war

It sits between $V_{T0,n}$ and $V_{DD} + V_{T0,p}$. The stronger side drags V_{th} towards the *other* side's threshold.

3 + 5 = 8

Symmetric: $V_{IL} = \frac{3V_{DD}+2V_T}{8}$, $V_{IH} = \frac{5V_{DD}-2V_T}{8}$; they add to V_{DD}.

Delay

"C over kV, times [sat bit + log bit]". Sat bit $2V_T/(V_{DD} - V_T)$; log bit $\ln(4 \cdot \text{overdrive}/V_{DD} - 1)$.

Flashcards (cover the answer)

Point B: which device is saturated?

The NMOS (the PMOS is linear).

Why is V_{OL} exactly 0?

The PMOS is off, so no current, so the linear NMOS has $V_{DS} = 0$.

Which sign after the square root for V_{th}?

Minus, because $V_{GS,p} - V_{T0,p} < 0$.

$k_R > 1$ moves the VTC…

Left (lower V_{th}).

$(W/L)_p$ for a symmetric inverter?

$\mu_n/\mu_p \approx 580/230 \approx 2.5$ times $(W/L)_n$.

$V_{IL} + V_{IH}$ (symmetric)?

V_{DD}.

Is C_{gs} part of C_{load}?

No. It loads the input node.

NMOS region at $V_{out} = V_{50\%}$ during the fall?

Linear (if $V_{50\%} < V_{OH} - V_{T0,n}$).

$t_1' - t_1$?

$2C_{load}V_{T0,n}/[k_n(V_{OH} - V_{T0,n})^2]$.

Ring of N inverters: period?

$2N\tau_P$ (N odd).

Gain of an inverter at V_{th}?

$-(g_{m1} + g_{m2})(r_{O1} \| r_{O2})$.

Energy per $0 \to 1 \to 0$ cycle?

$C_{load}V_{DD}^2$, half in each transistor.

Exam playbook

1. **Any CMOS VTC question:** compute $k_n = \mu_n C_{ox}(W/L)_n$, k_p and k_R first. Write the four terminal voltages, then V_{th} from the boxed formula.

2. **V_{IL} / V_{IH}:** write ① or ③, differentiate, set slope $= -1$, solve for V_{out} in terms of V_{IL} (or V_{IH}), substitute, solve the quadratic. Check with the ρ shortcut. If symmetric, use the $3/8$ and $5/8$ forms directly.

3. **Design for V_{th}:** $k_R = \left(\frac{V_{DD}+V_{T0,p}-V_{th}}{V_{th}-V_{T0,n}}\right)^2$, then convert to W/L with the mobility ratio. "Equal rise/fall" means symmetric.

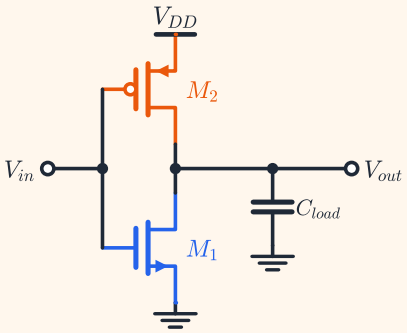
4. **Delay:** write C_{load}, decide each transistor's region at both endpoints, then use average current or the exact formula, whichever the question asks for. Always state $V_{50\%}$.

TOP 5 MISTAKES

1. Using the saturation formula at both ends of the average-current method.
2. Taking the $+$ sign after the square root in the V_{th} derivation.
3. Forgetting $\frac{dV_{out}}{dV_{in}}$ on the V_{out} terms when differentiating.
4. Mixing μC_{ox} with k: always multiply by W/L first.
5. Counting C_{gs} in the load, or forgetting the next stage's gate.

Capstone problem (all three lectures)

ONE INVERTER, START TO FINISH



$V_{DD} = 2.5$ V, $V_{T0,n} = 0.5$, $V_{T0,p} = -0.5$ V
$k_n = 300$ µA/V², $k_p = 150$ µA/V²
$C_{load} = 50$ fF

(a) V_{OH}, V_{OL} and V_{th}. (b) V_{IL}, V_{IH} and both noise margins. (c) τ_{PHL} and τ_{PLH} by the exact formulas, and τ_P. (d) Check τ_{PHL} with the average-current method. (e) How would you make the edges equal, and what happens to V_{th}?

Solution

(a) $V_{OH} = 2.5$ V, $V_{OL} = 0$. $k_R = 2$, $\sqrt{1/k_R} = 0.707$: $V_{th} = \dfrac{0.5 + 0.707 \times 2.0}{1.707} = \mathbf{1.121\ V}$.

(b) ② gives $V_{out} = \frac{1}{2}(3V_{IL} + 2.0)$. Substitute into ①: $V_{IL} = 0.897$ **V** ($V_{out} = 2.346$ V).
④ gives $3V_{IH} = 2.0 + 2(2V_{out} + 0.5)$. Substitute into ③: $V_{IH} = 1.268$ **V** ($V_{out} = 0.201$ V).
So $NM_L = 0.897$ **V and** $NM_H = 2.5 - 1.268 = 1.232$ **V**.
Shortcut check: $\rho_{IL} = 0.360$ gives $V_{IL} = (0.5 + 0.360 \times 2.0)/1.360 = 0.897$ V ✓.

(c) Both devices have $|V_T| = 0.5$ V, so both brackets are the same:

$$\left[\frac{2(0.5)}{2.0} + \ln\left(\frac{4(2.0)}{2.5} - 1\right)\right] = 0.5 + 0.789 = 1.289$$

$$\tau_{PHL} = \frac{50\text{f}}{300\mu \times 2.0} \times 1.289 = 83.3\ \text{ps} \times 1.289 = \mathbf{107\ ps}$$

$$\tau_{PLH} = \frac{50\text{f}}{150\mu \times 2.0} \times 1.289 = 166.7\ \text{ps} \times 1.289 = \mathbf{215\ ps}$$

$\tau_P = \frac{1}{2}(107 + 215) = \mathbf{161\ ps}$. The PMOS is half as strong, so the rising edge is exactly twice as slow.

(d) At $V_{out} = 2.5$ V the NMOS is saturated: $150\mu(2.0)^2 = 600\ \mu$A. At $V_{out} = 1.25$ V it is linear: $150\mu[2(2.0)(1.25) - 1.25^2] = 516\ \mu$A. The average is $558\ \mu$A, so $\tau_{PHL} \approx 50\text{f} \times 1.25/558\mu = \mathbf{112\ ps}$ (4 % above the exact value, as expected).

(e) Make $k_p = k_n$: double $(W/L)_p$. Then τ_{PLH} drops to about 107 ps (if C_{load} stays the same; in reality the wider PMOS adds some capacitance). Now $k_R = 1$ with matched thresholds, so the inverter becomes symmetric: V_{th} moves right to $V_{DD}/2 = 1.25$ **V** and the noise margins become equal ($V_{IL} = (7.5 + 1.0)/8 = 1.0625$ V).
